MOTOR VIBRATION EXPERIMENTAL DATA ANALYSIS

Dataset	adxl345_5sec_calibrated
Sensor	ADXL345
Controller	ESP32
Sampling Frequency	1559.00 Hz
Number of Samples	49930
Duration	32.027 s

1. Dataset Information

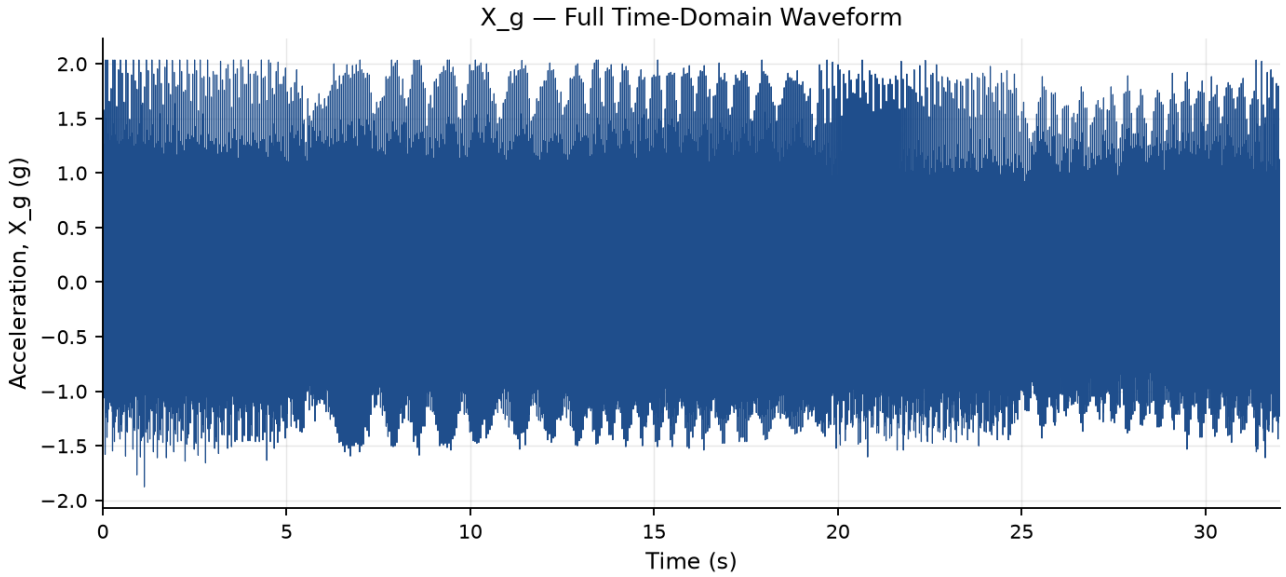
Parameter	Value
Input file	adxl345_5sec_calibrated.csv
Samples	49930
Sampling frequency	1559.00 Hz
Duration	32.027 s
Nyquist frequency	779.50 Hz
Frequency resolution	0.0312 Hz
Sensor	ADXL345
Controller	ESP32
Axes	X, Y, Z
Units	g (gravitational acceleration)

2. Signal Processing Methodology

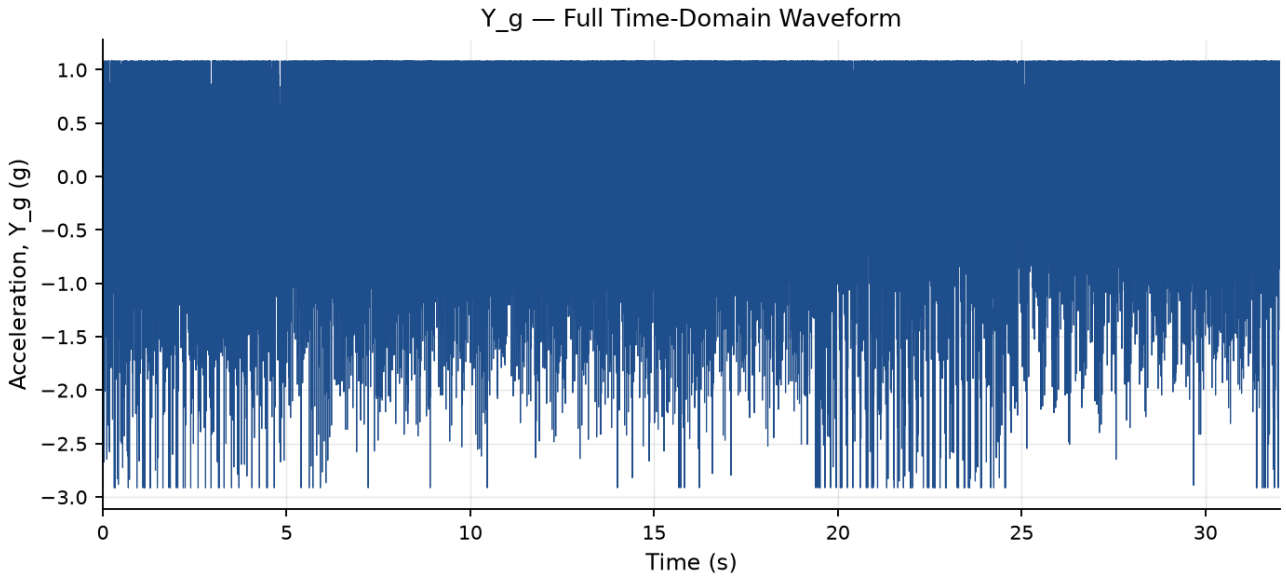
Vibration data were acquired with an ADXL345 accelerometer connected to an ESP32, sampled at 1559.0 Hz. The input CSV is already calibrated/processed and contains acceleration channels X_g, Y_g, Z_g. The mean (DC) component of each channel was removed to obtain AC vibration signals X_ac, Y_ac, Z_ac. A resultant vibration signal was formed as $R_g = \sqrt{X_{ac}^2 + Y_{ac}^2 + Z_{ac}^2}$. Frequency-domain analysis applies a Hann window before computing a single-sided amplitude-spectrum FFT (window-gain corrected) over the complete 0–Nyquist range. Power Spectral Density is estimated with Welch's method, and time–frequency content is examined with a short-time Fourier spectrogram. Significant spectral peaks are detected with an adaptive prominence threshold derived from each dataset's own robust noise floor (median and median-absolute-deviation of the power spectrum), so the detection automatically scales with the amplitude of the current dataset rather than relying on a fixed value. All numerical results below are directly measured quantities; the report performs no automatic engineering interpretation or fault classification.

3. Time-Domain Waveforms (Processed Signals)

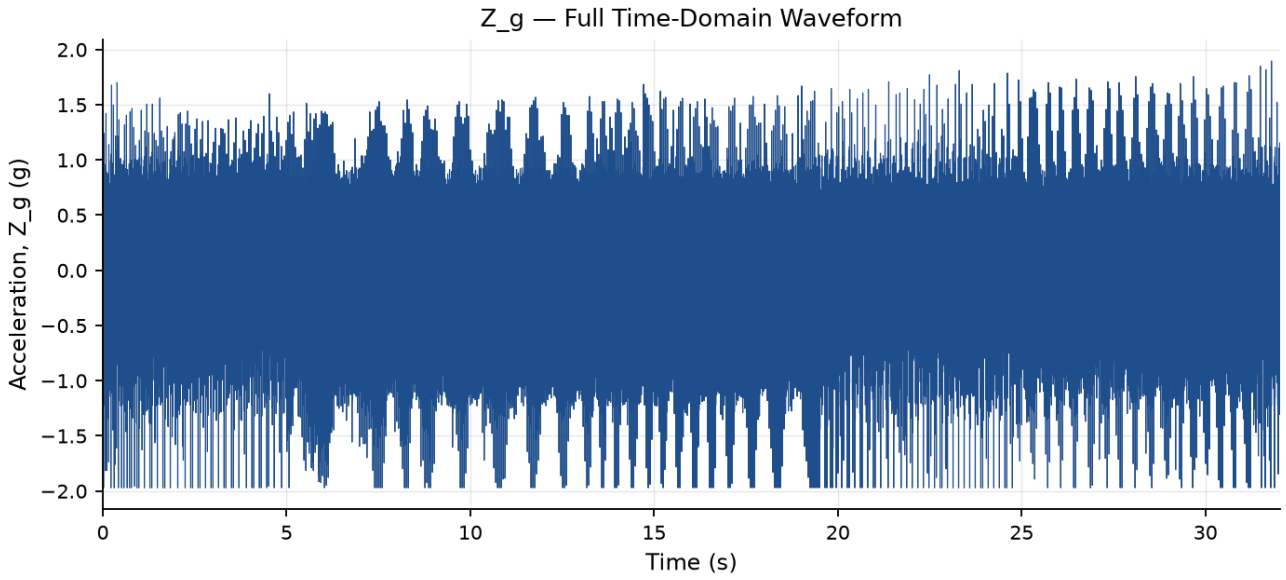
Full-duration waveforms of the processed acceleration channels, followed by a 3-second zoom window for waveform-shape inspection.



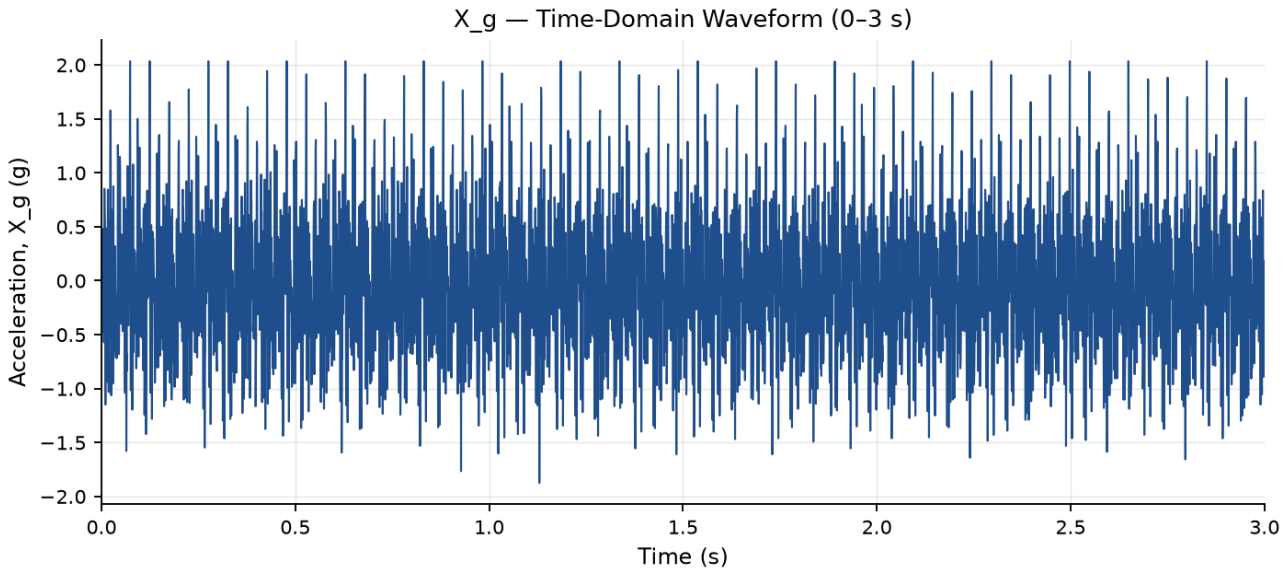
X_g — full time-domain waveform.



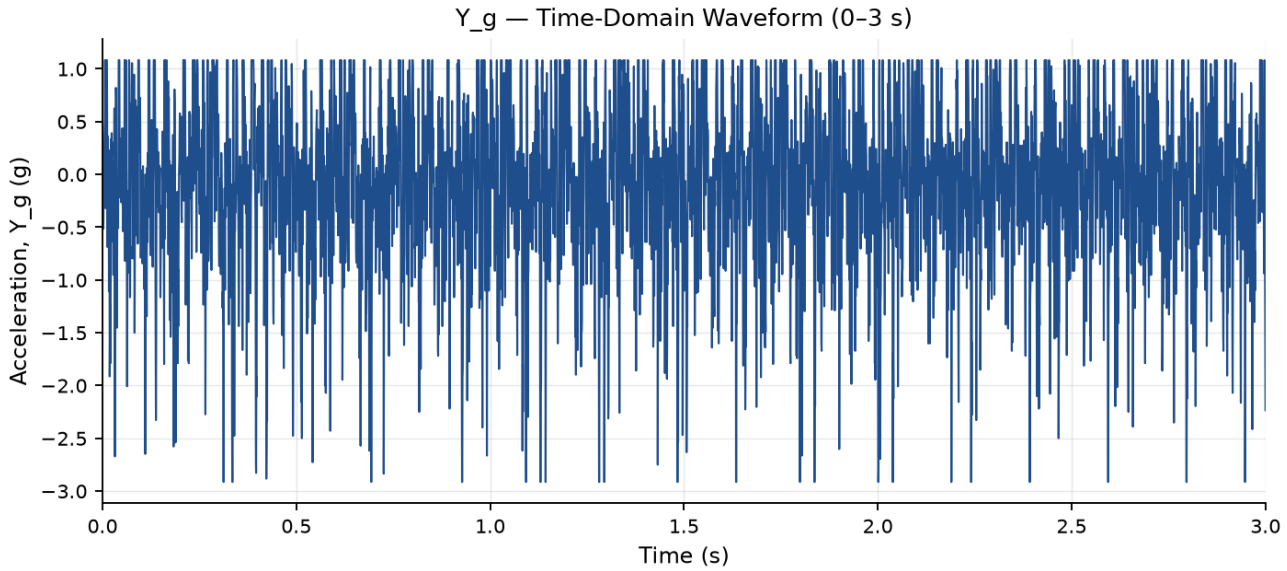
Y_g — full time-domain waveform.



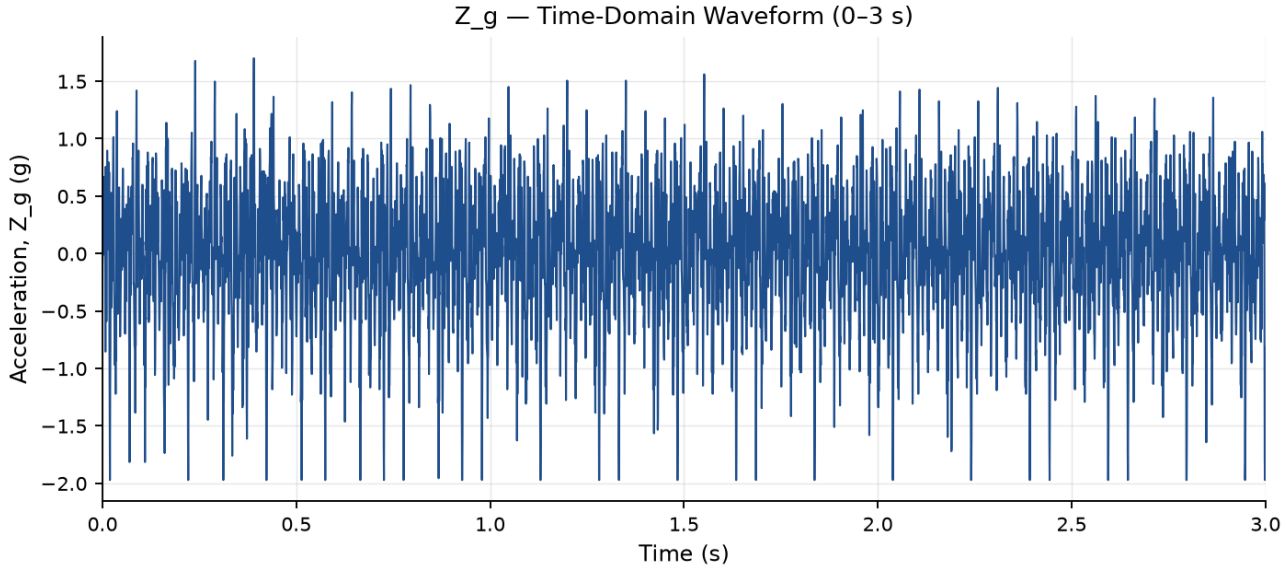
Z_g — full time-domain waveform.



X_g — zoomed waveform (0–3 s).



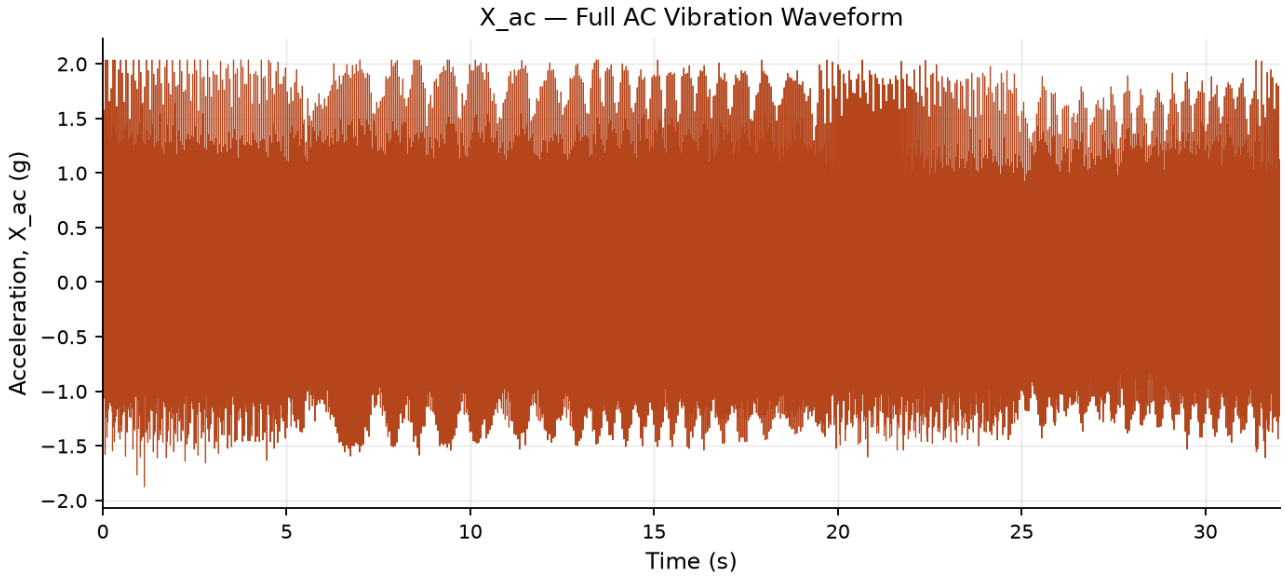
Y_g — zoomed waveform (0–3 s).



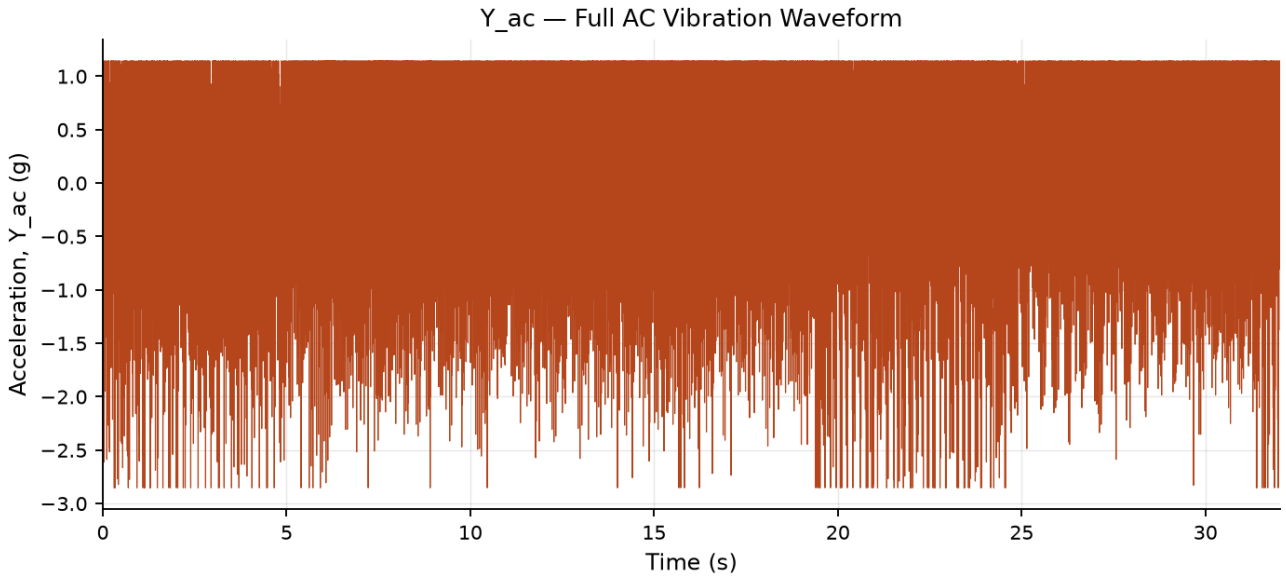
Z_g — zoomed waveform (0–3 s).

4. AC Vibration Waveforms

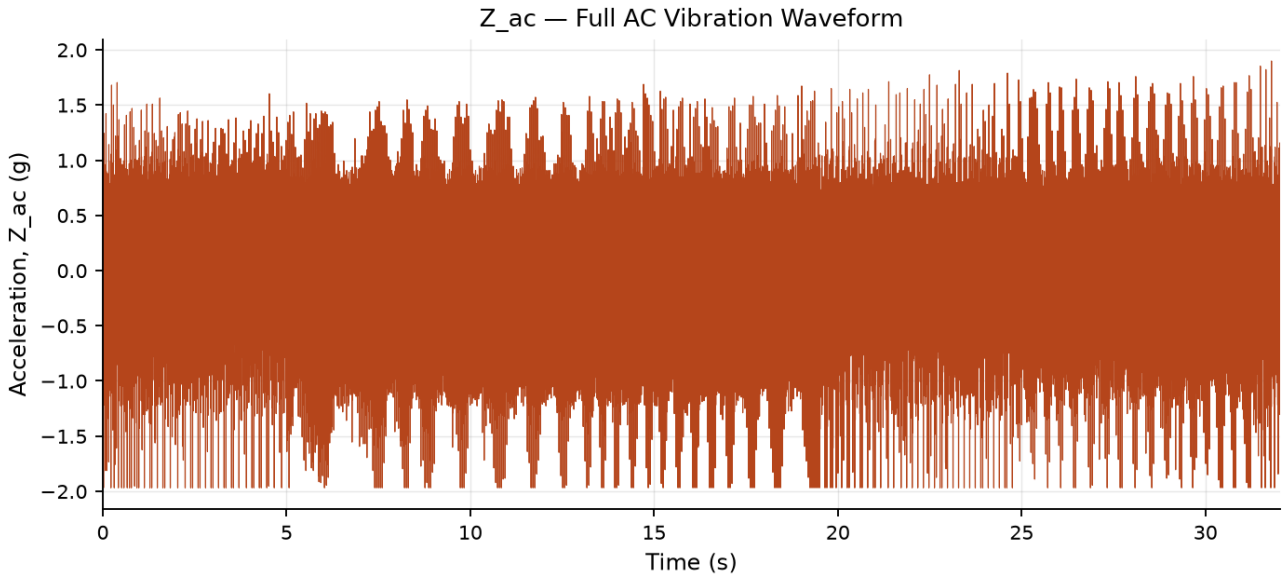
DC-removed AC vibration signals (X_ac, Y_ac, Z_ac) and the resultant vibration signal R_g = sqrt(X_ac²+Y_ac²+Z_ac²). Full duration and zoomed views.



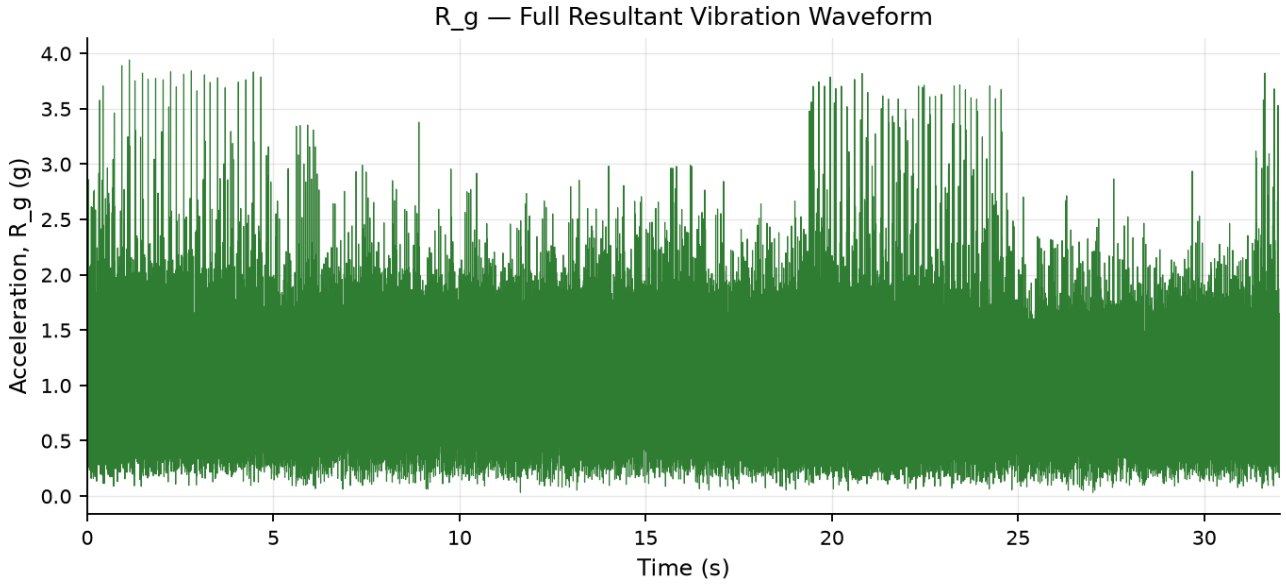
X_ac — full waveform.



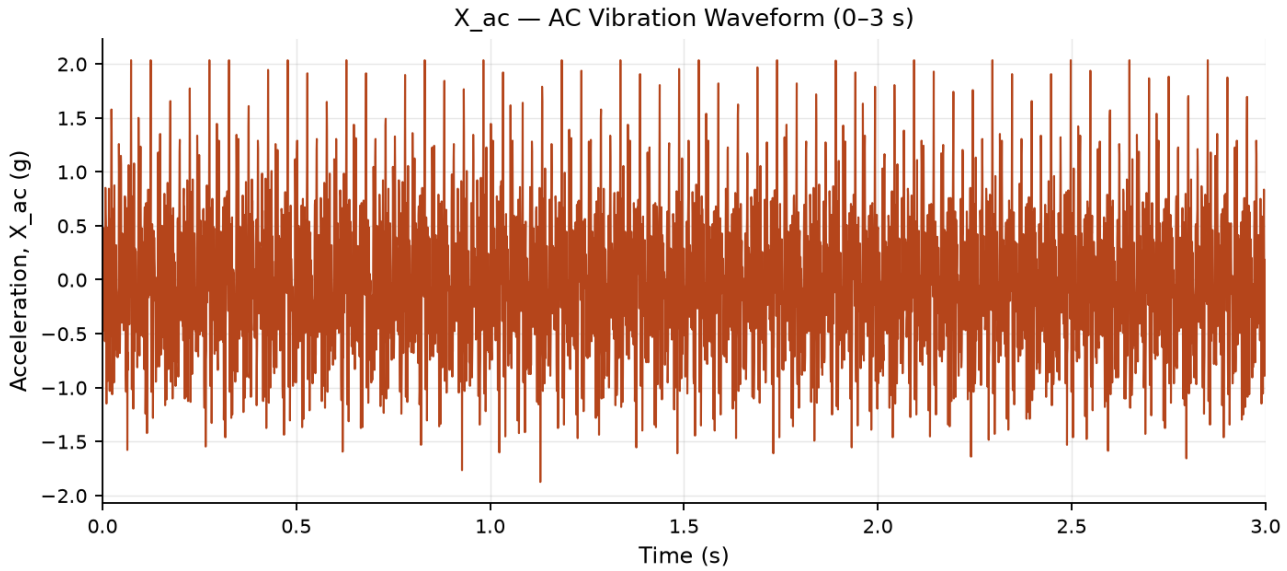
Y_ac — full waveform.



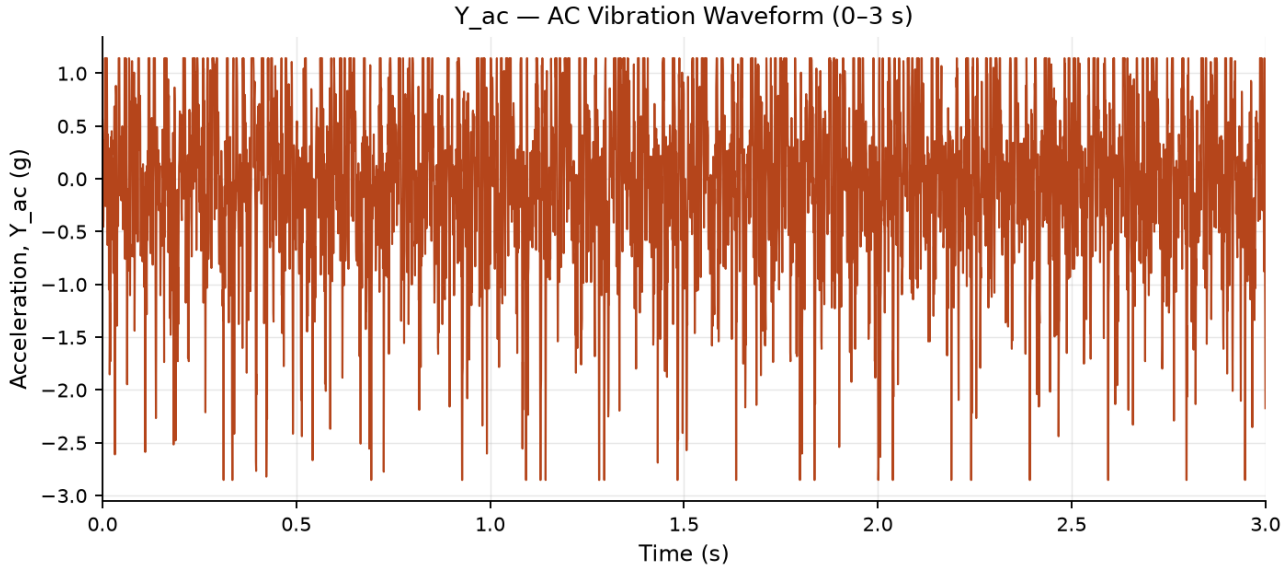
Z_ac — full waveform.



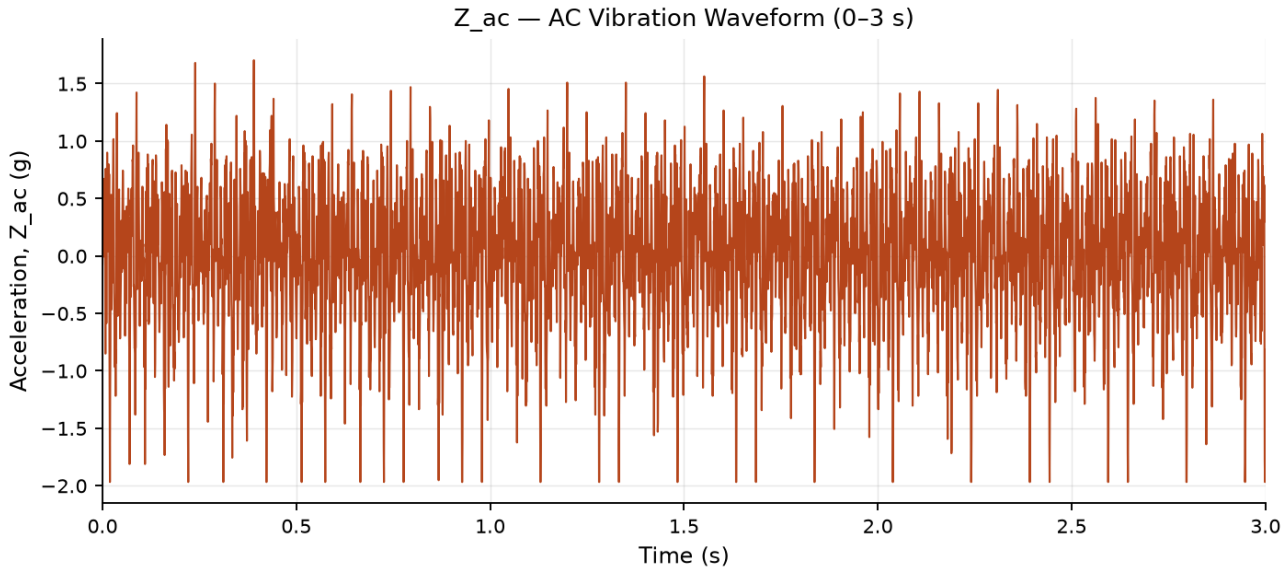
R_g — full waveform.



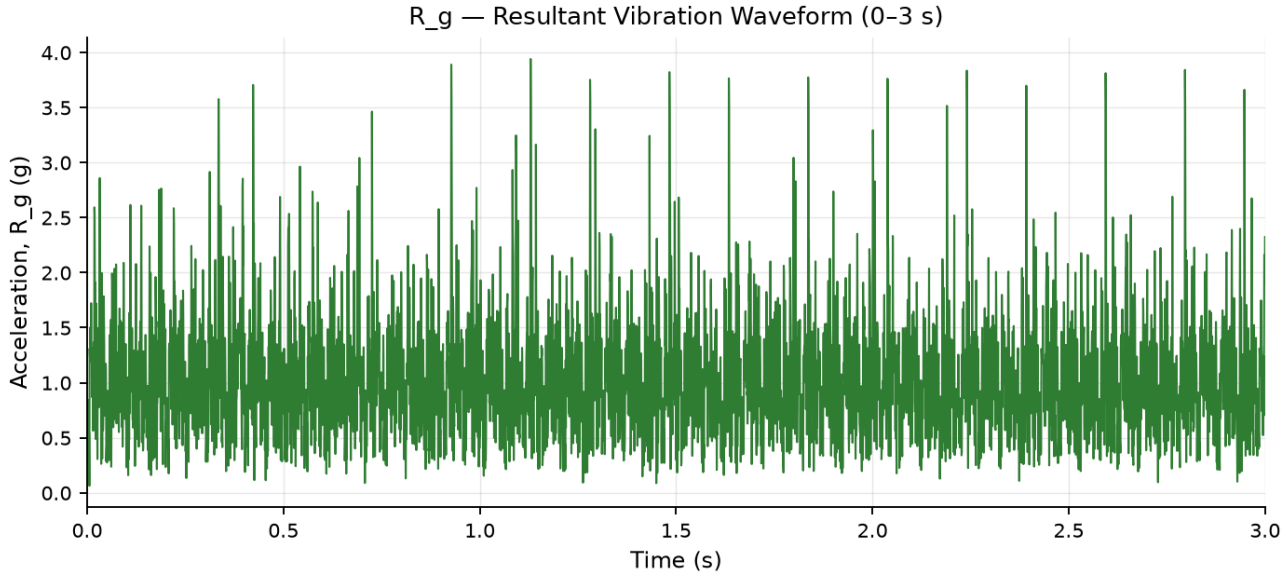
X_ac — zoomed waveform (0–3 s).



Y_ac — zoomed waveform (0–3 s).



Z_ac — zoomed waveform (0–3 s).



R_g — zoomed waveform (0–3 s).

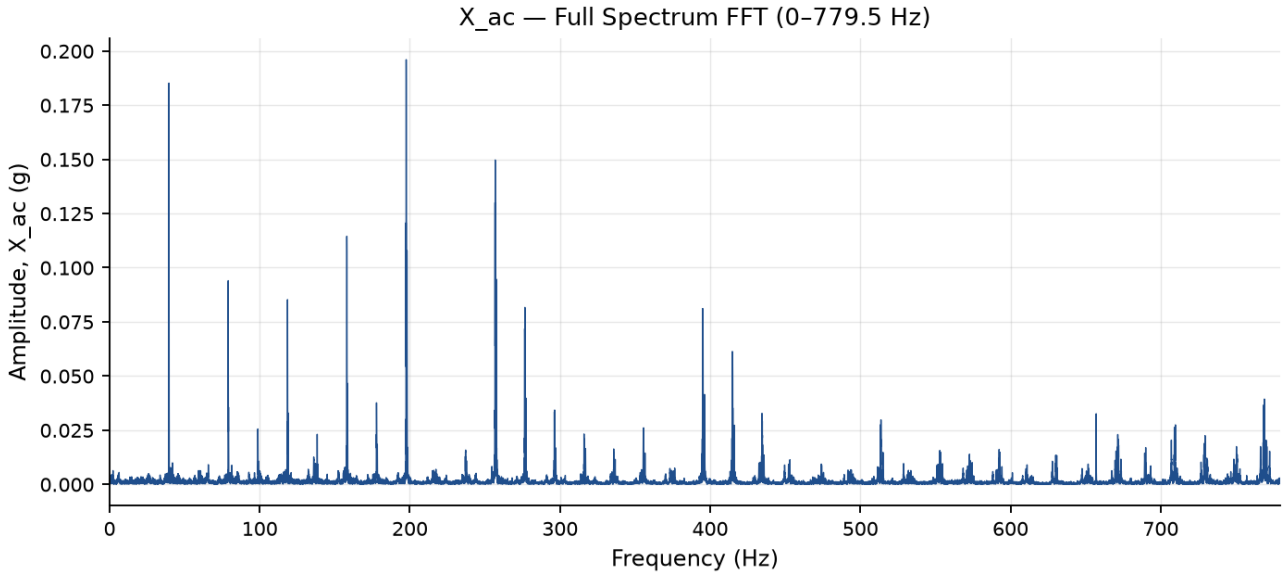
5. Time-Domain Feature Results

Time-domain statistical and shape features computed on X_ac, Y_ac, Z_ac and R_g. Each value is reported once and is not repeated elsewhere in this report.

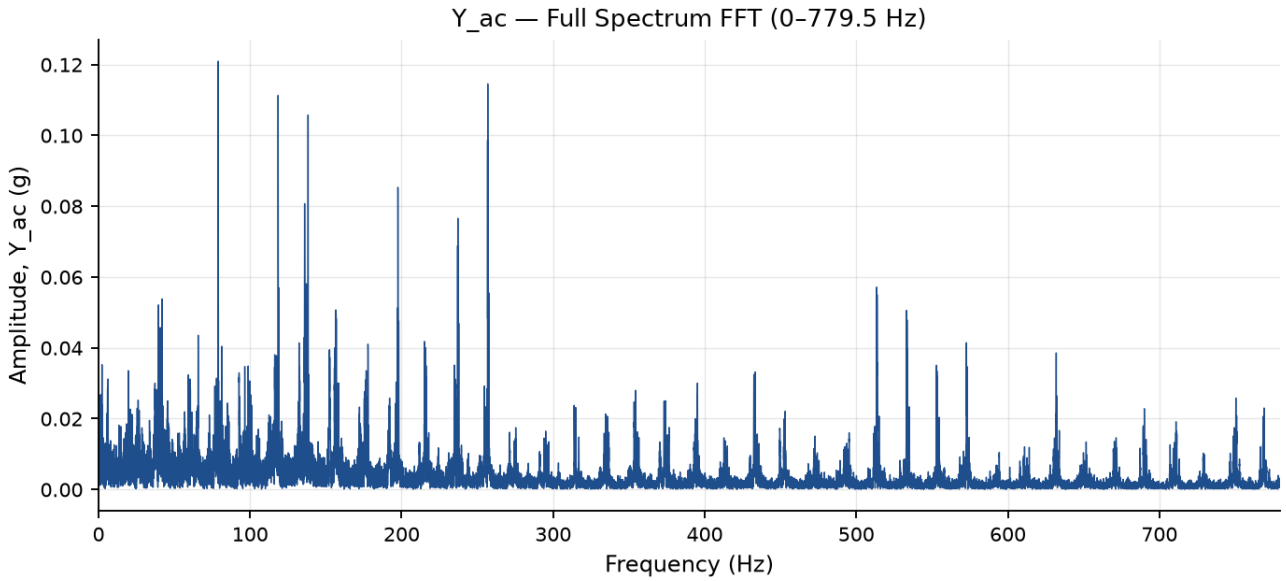
Feature	X_ac	Y_ac	Z_ac	R_g
Mean	1.0709e-17	-6.2615e-18	5.3650e-17	0.9609
Median	-0.023537	0.0058614	0.047371	0.9081
Std Dev	0.57812	0.7111	0.55299	0.47154
Variance	0.33422	0.50567	0.30579	0.22235
RMS	0.57812	0.7111	0.55299	1.0704
Maximum	2.0351	1.1426	1.8989	3.9411
Minimum	-1.8751	-2.8535	-1.9683	0.032901
Absolute Peak	2.0351	2.8535	1.9683	3.9411
Peak to Peak	3.9102	3.9961	3.8672	3.9082
Crest Factor	3.5201	4.0128	3.5593	3.682
Shape Factor	1.2842	1.3085	1.2822	1.1139
Impulse Factor	4.5207	5.2508	4.5639	4.1015
Clearance Factor	5.4101	6.4022	5.4855	4.3617
Skewness	0.30781	-0.52344	-0.47875	1.0389
Kurtosis	0.44415	0.81082	0.44784	2.6373
Mean Absolute Value	0.45016	0.54344	0.43127	0.9609
Energy	16688	25248	15268	57204
Signal Power	0.33422	0.50567	0.30579	1.1457
Range	3.9102	3.9961	3.8672	3.9082
MAD	0.35938	0.41406	0.35156	0.31176
Q1	-0.36729	-0.38476	-0.33544	0.61814
Q3	0.35146	0.44336	0.3755	1.2382
IQR	0.71875	0.82812	0.71094	0.62011
Zero Crossing Rate	0.3502	0.31479	0.27978	0

6. Frequency-Domain Analysis (FFT)

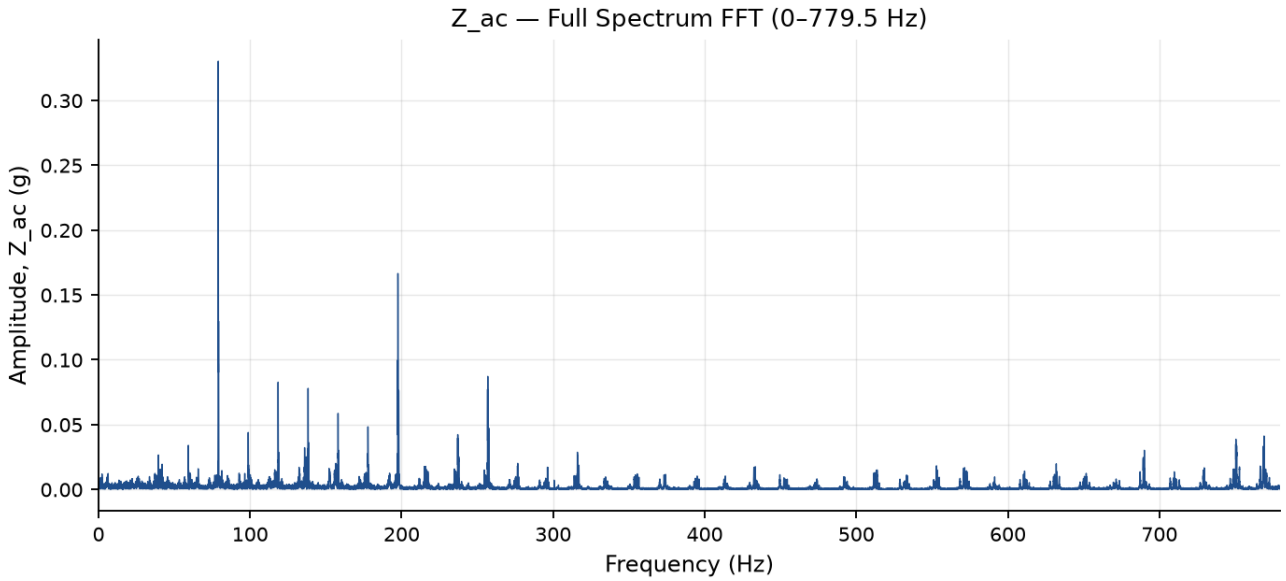
Single-sided amplitude spectra (Hann-windowed) over the complete measured range, 0–779.5 Hz. The 100 Hz zoom below is a visualization aid only and is not an analysis boundary; full-spectrum peak detection was performed over the entire range (see Section 13).



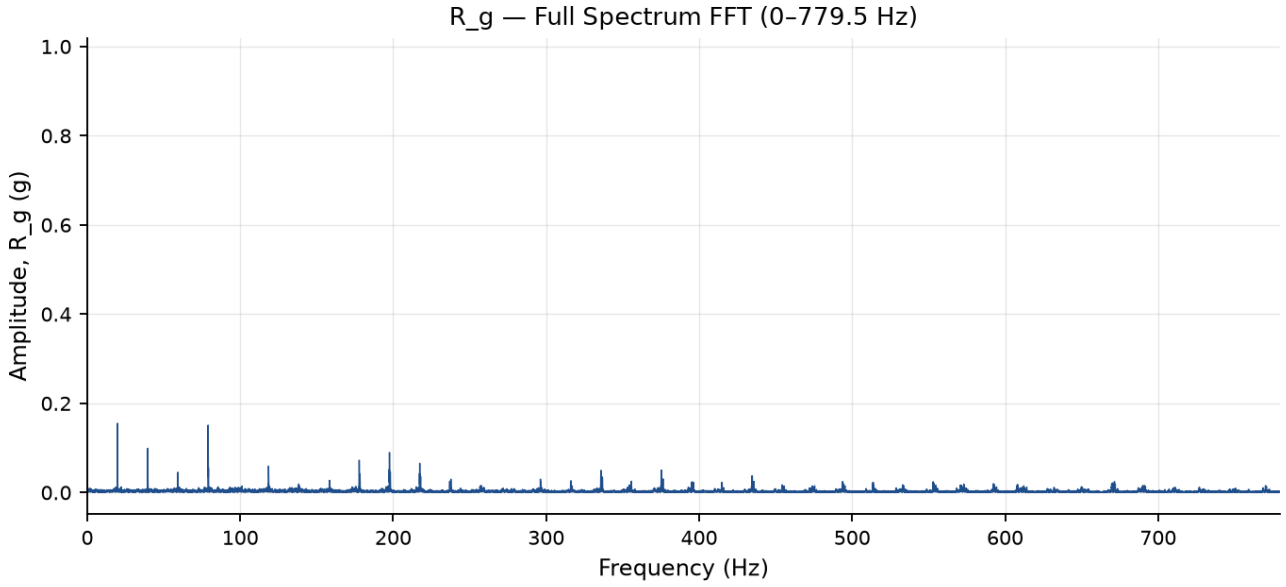
X_ac — full spectrum FFT, 0–779.5 Hz.



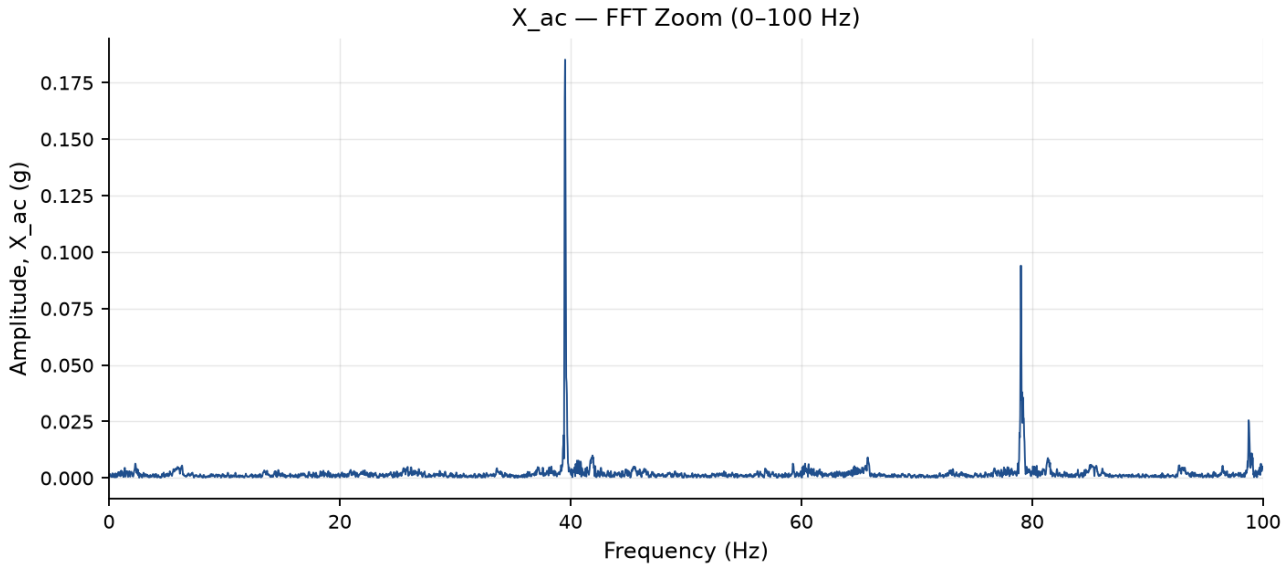
Y_ac — full spectrum FFT, 0–779.5 Hz.



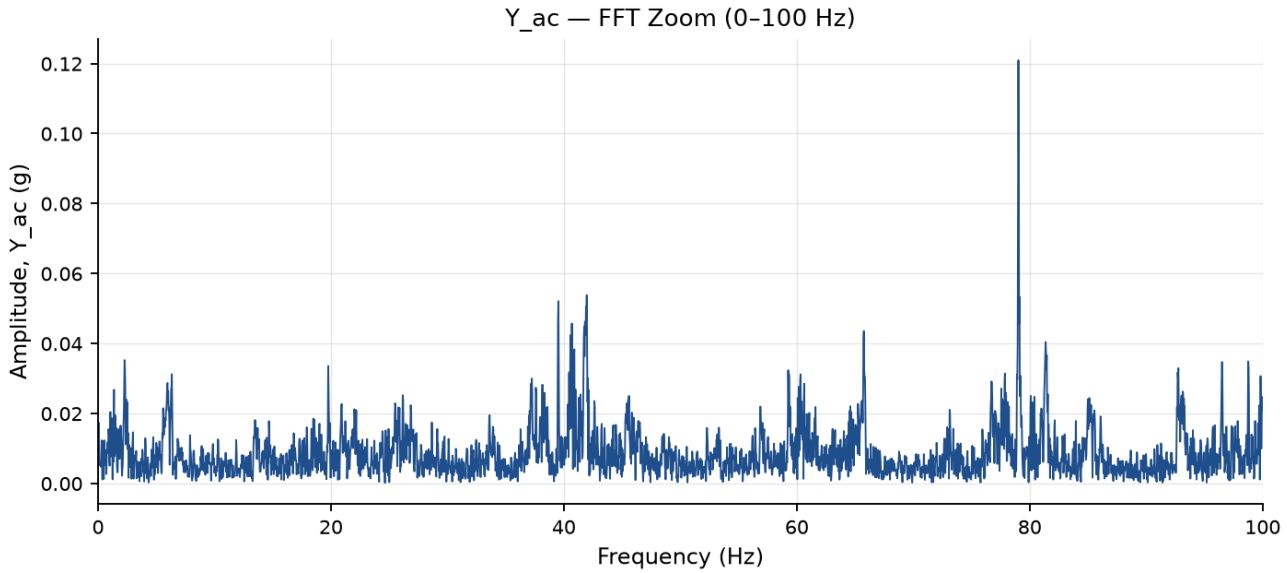
Z_ac — full spectrum FFT, 0–779.5 Hz.



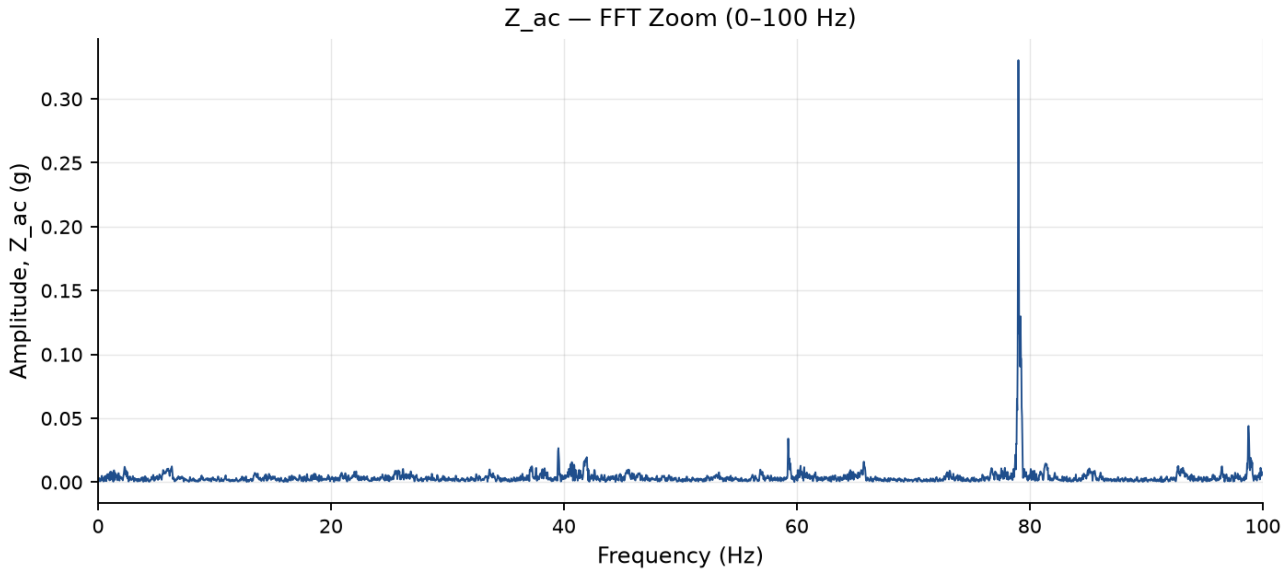
R_g — full spectrum FFT, 0–779.5 Hz.



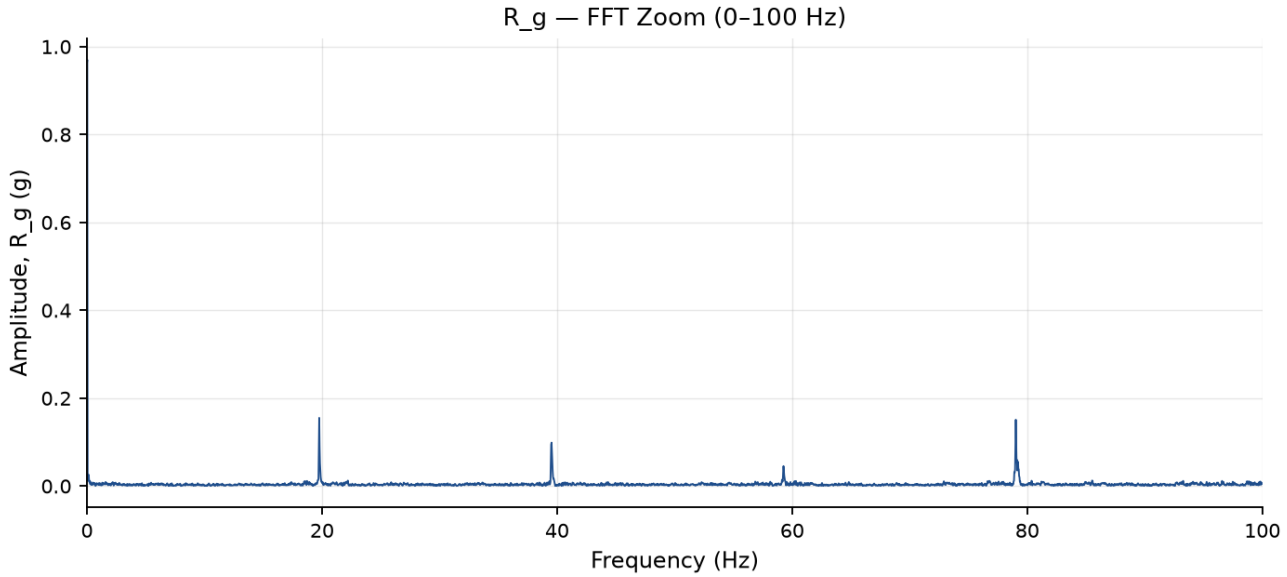
X_ac — FFT, 0–100 Hz zoom (visualization only).



Y_ac — FFT, 0–100 Hz zoom (visualization only).



Z_ac — FFT, 0–100 Hz zoom (visualization only).



R_g — FFT, 0–100 Hz zoom (visualization only).

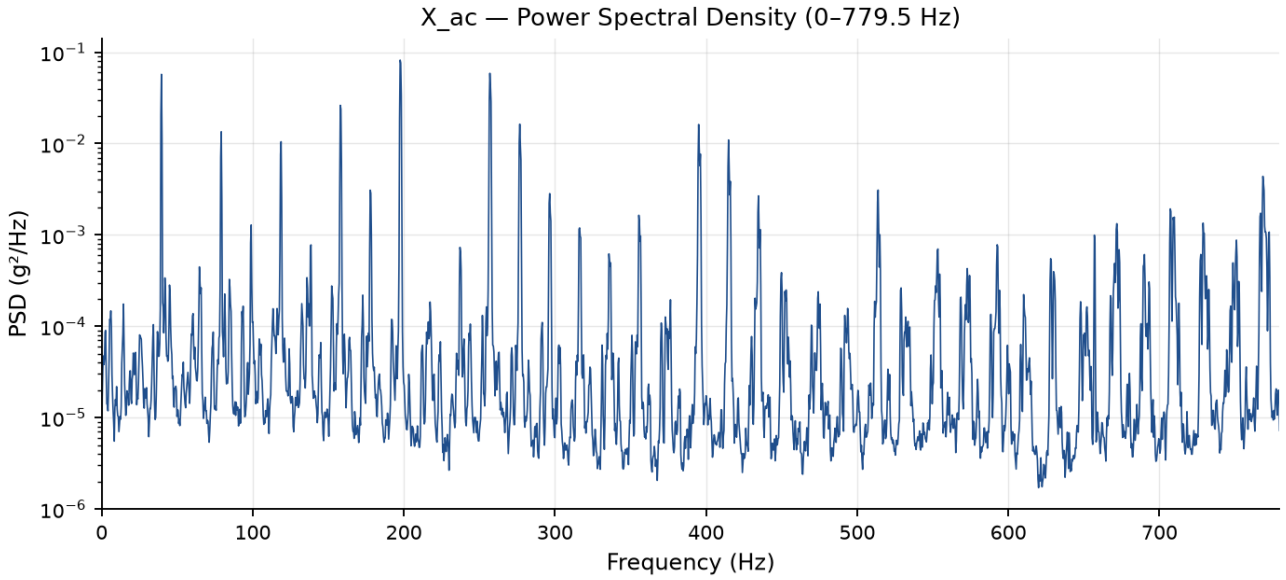
7. Frequency-Domain Feature Results

Spectral features computed from the full 0–Nyquist amplitude/power spectrum (DC excluded). Reported once; not repeated elsewhere in this report.

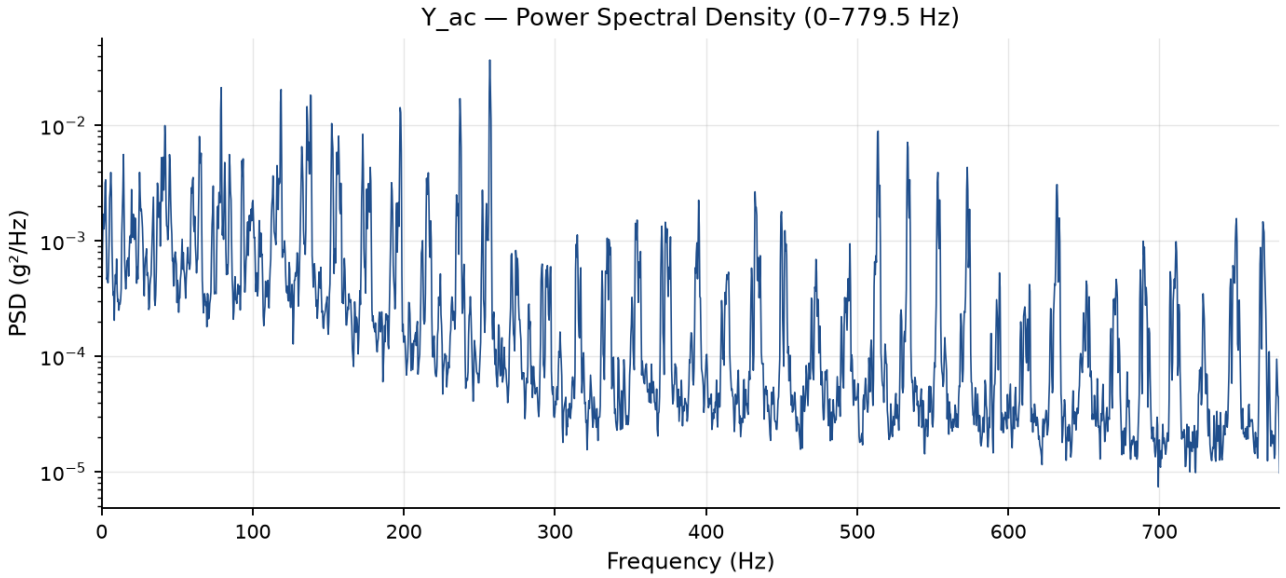
Feature	X_ac	Y_ac	Z_ac	R_g
Dominant Frequency Hz	197.55	79.027	79.027	0.031224
Dominant Amplitude g	0.19602	0.12094	0.33027	0.96968
Spectral Centroid Hz	264.91	216.92	204.85	106.65
Spectral RMS Frequency Hz	318.18	281.23	279.02	211.39
Spectral Bandwidth Hz	176.25	178.99	189.45	182.51
Spectral Energy	1.0068	1.5131	0.92685	1.5974
Spectral Variance	31062	32039	35891	33311
Spectral Skewness	1.293	1.2731	1.8767	1.8304
Spectral Kurtosis	4.4424	3.8466	5.5774	5.4477
Spectral Flatness	0.026588	0.10726	0.034049	0.066337
Spectral Entropy	0.5735	0.79206	0.57782	0.3717
Spectral Rolloff Hz	414.78	433.32	296.34	316.02
Spectral Peak Power	0.038422	0.014627	0.10908	0.94028
Number of Significant Peaks	5	5	5	5

8. Power Spectral Density

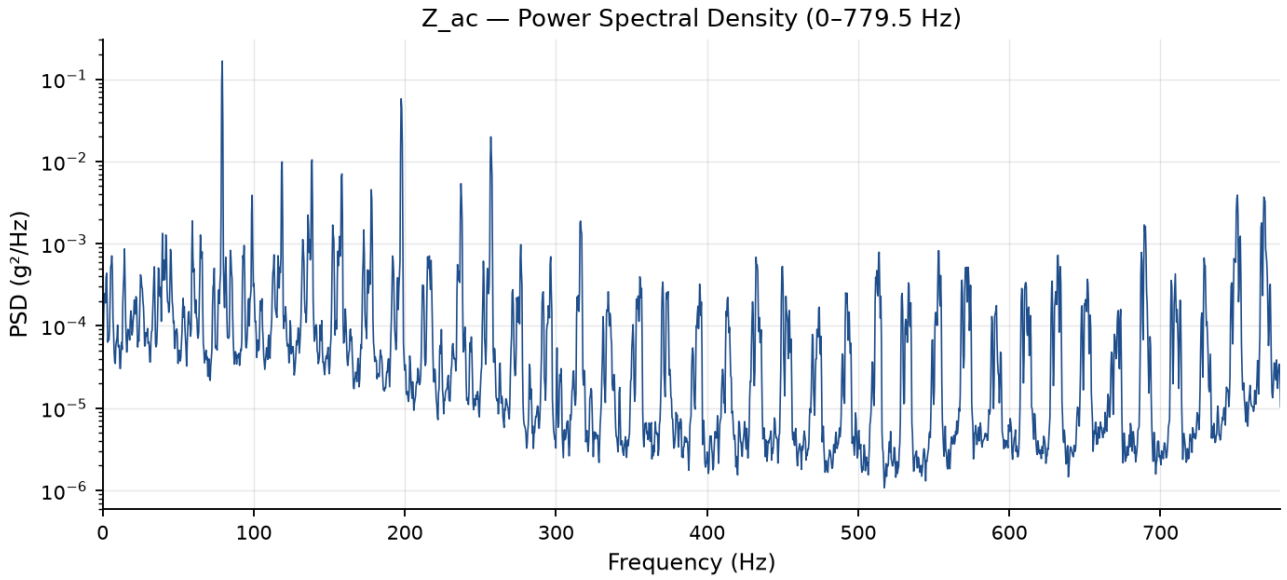
Welch PSD (window=hann, nperseg=4096, noverlap=2048) over the full valid range, followed by a 100 Hz visualization zoom. The full PSD is the primary complete-spectrum analysis.



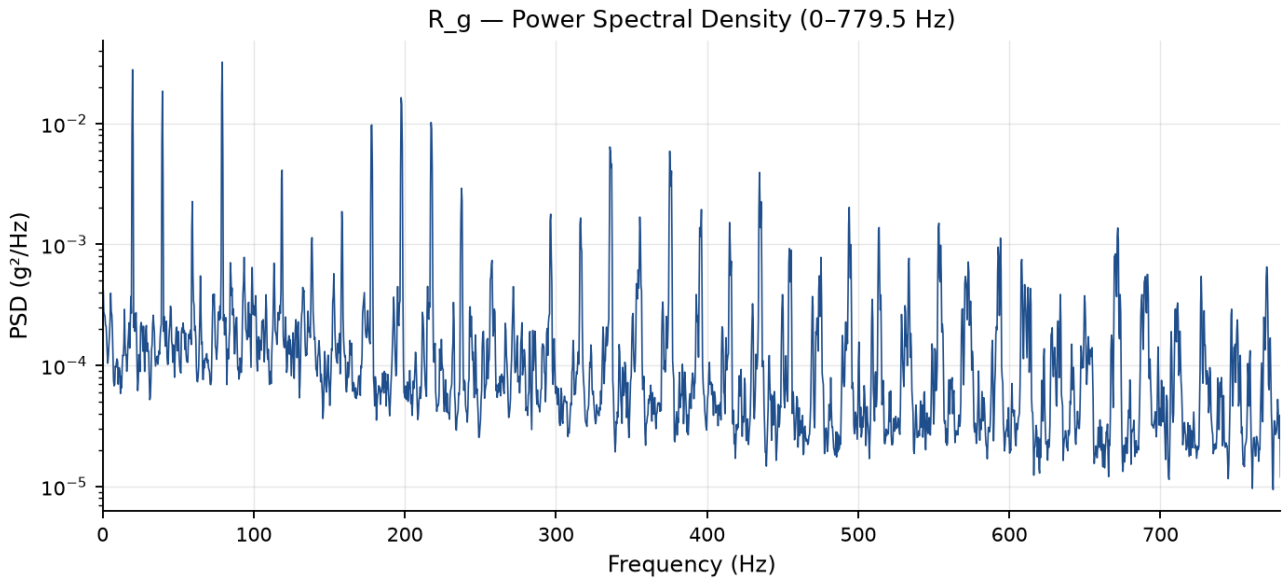
X_ac — full PSD, 0–779.5 Hz.



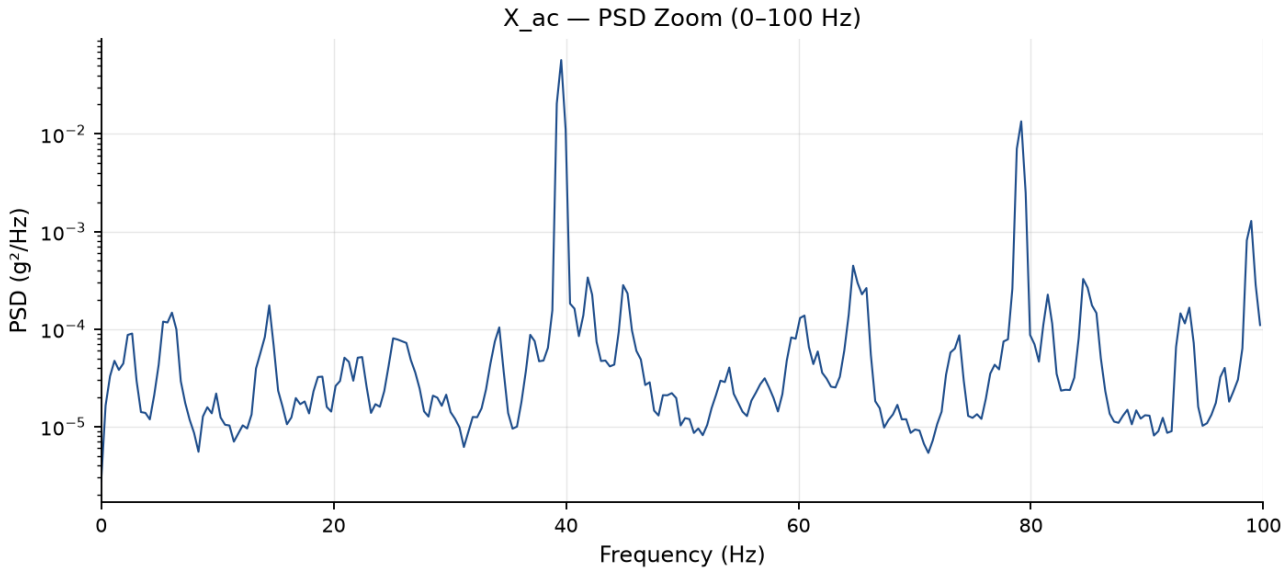
Y_ac — full PSD, 0–779.5 Hz.



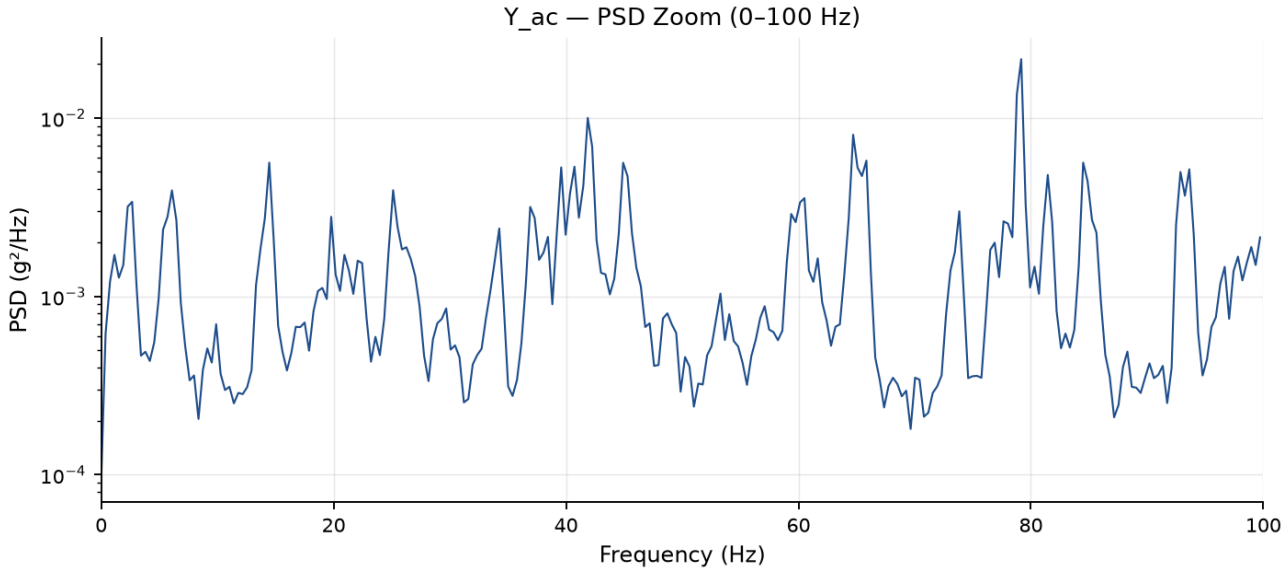
Z_ac — full PSD, 0–779.5 Hz.



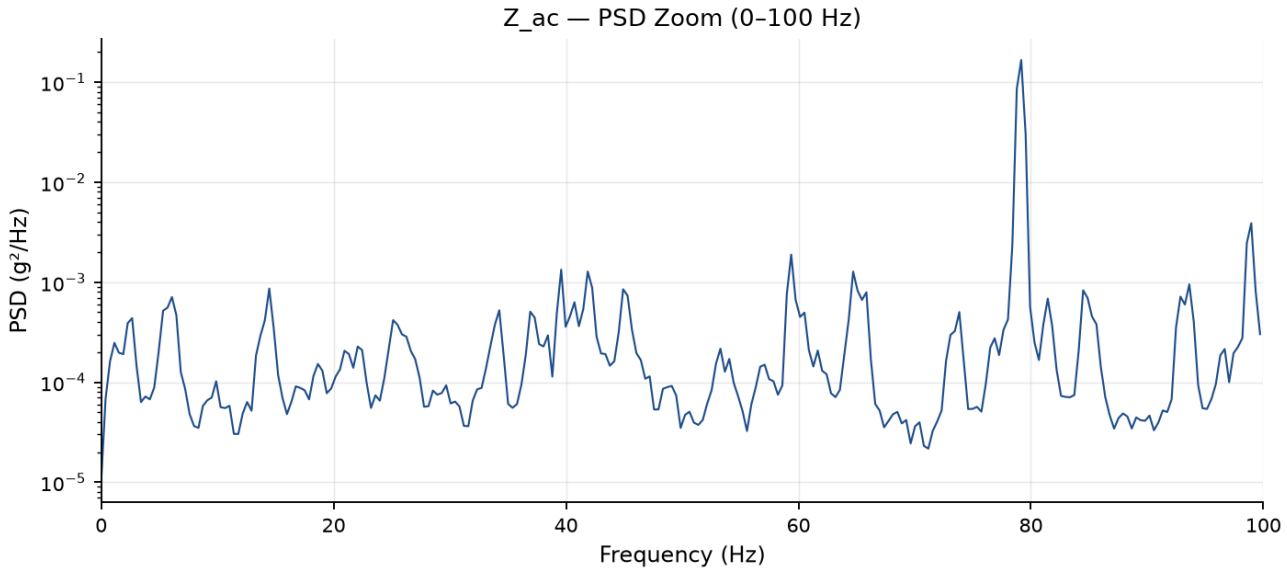
R_g — full PSD, 0–779.5 Hz.



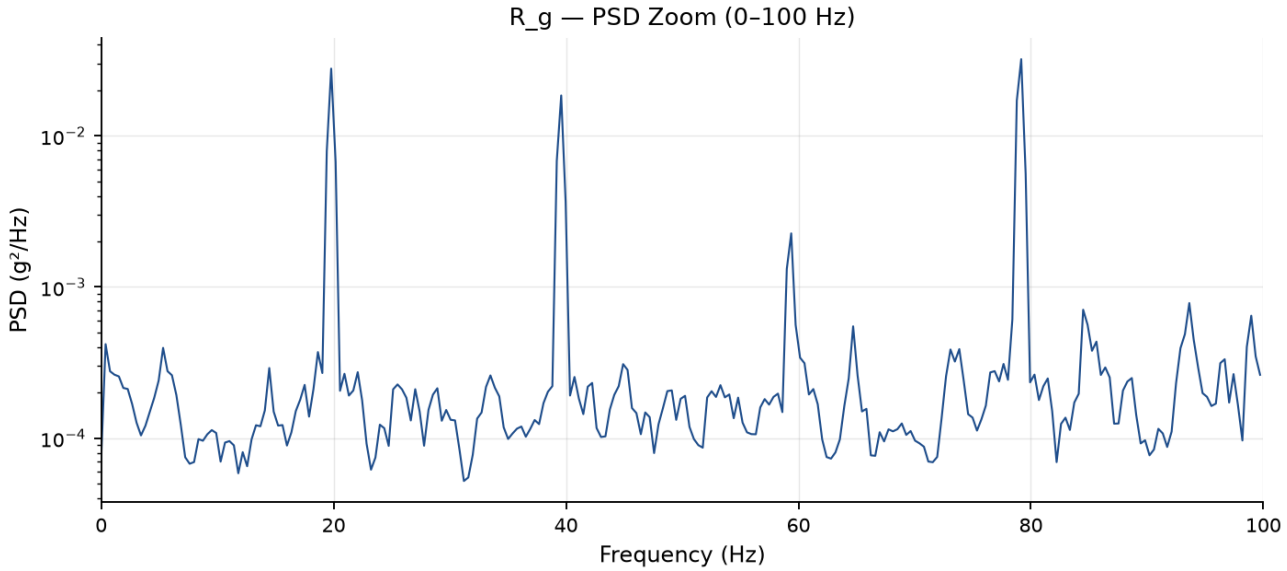
X_ac — PSD, 0–100 Hz zoom.



Y_ac — PSD, 0–100 Hz zoom.



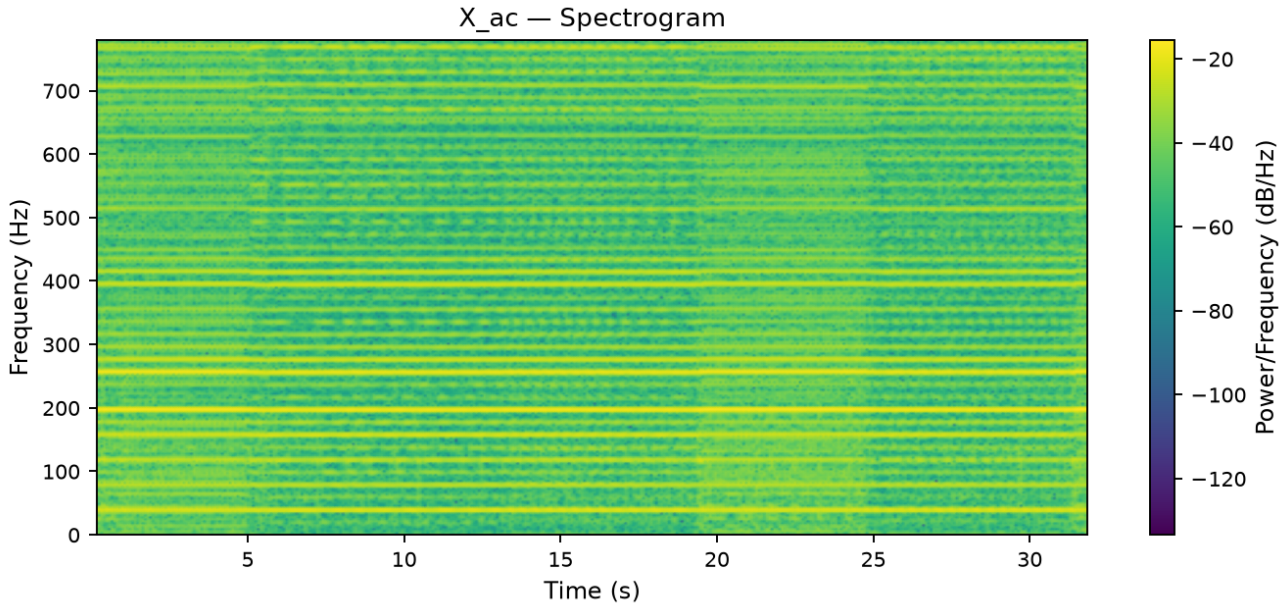
Z_ac — PSD, 0–100 Hz zoom.



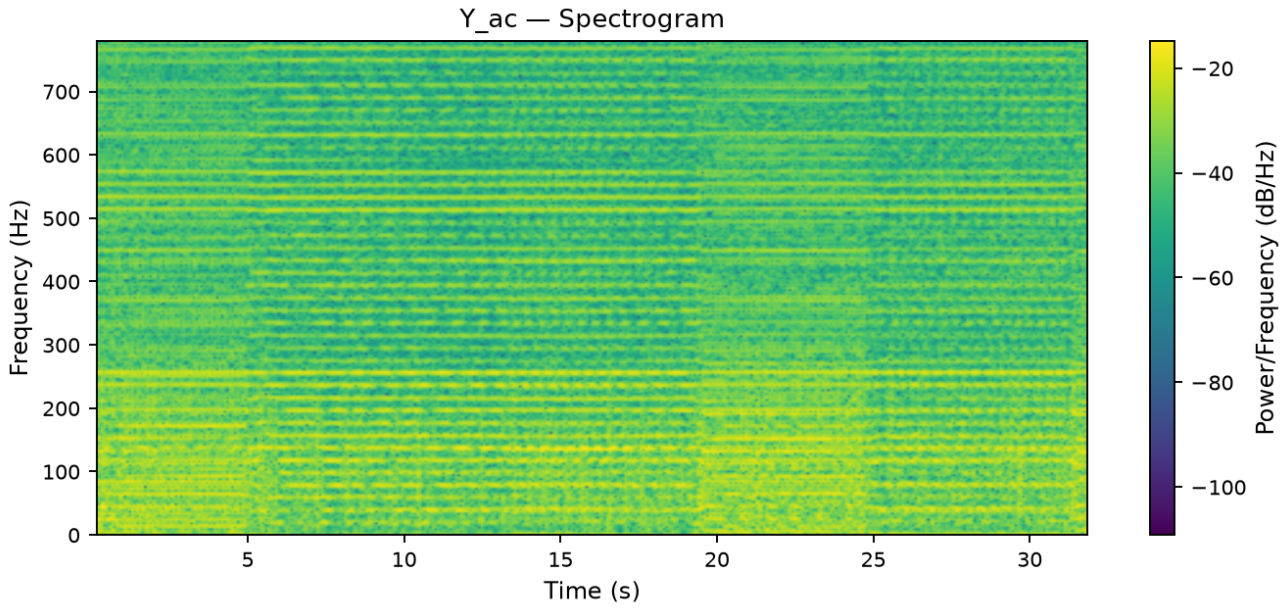
R_g — PSD, 0–100 Hz zoom.

9. Spectrogram Analysis

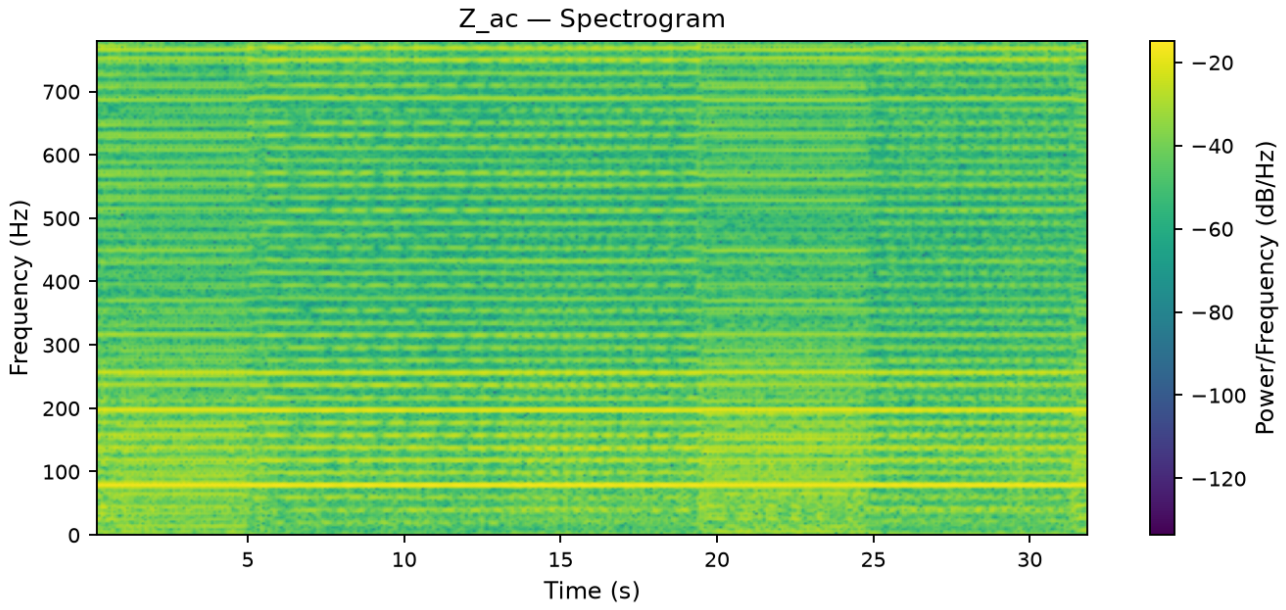
Short-time Fourier spectrograms (nperseg=512, noverlap=384) over the complete available frequency range.



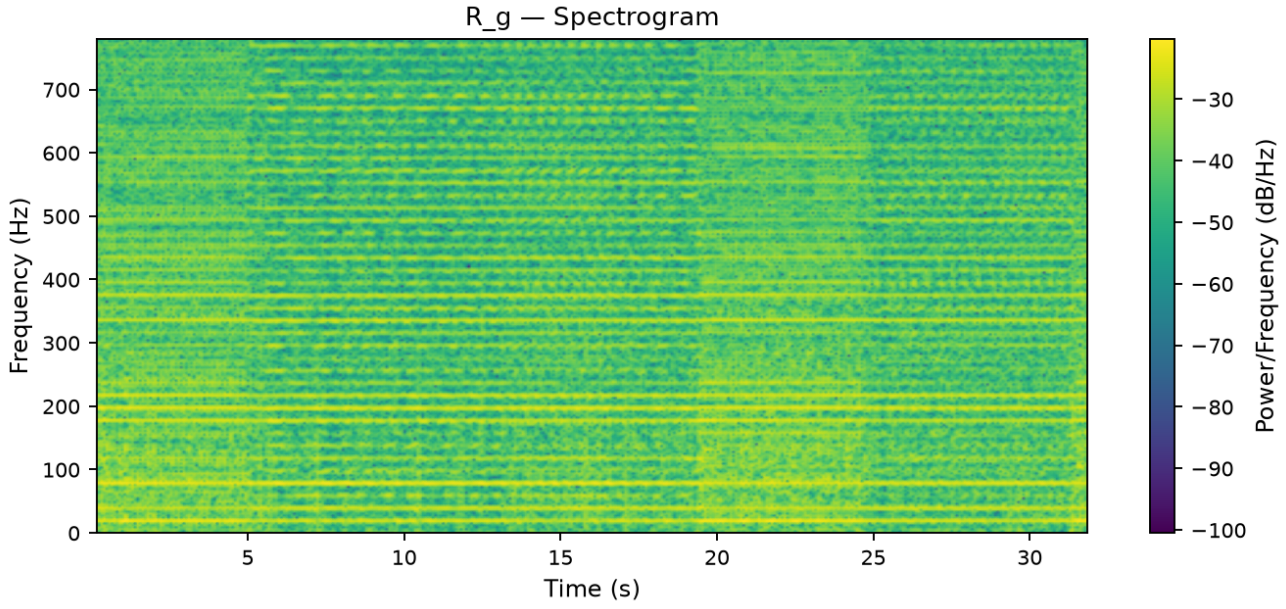
X_ac — spectrogram.



Y_ac — spectrogram.



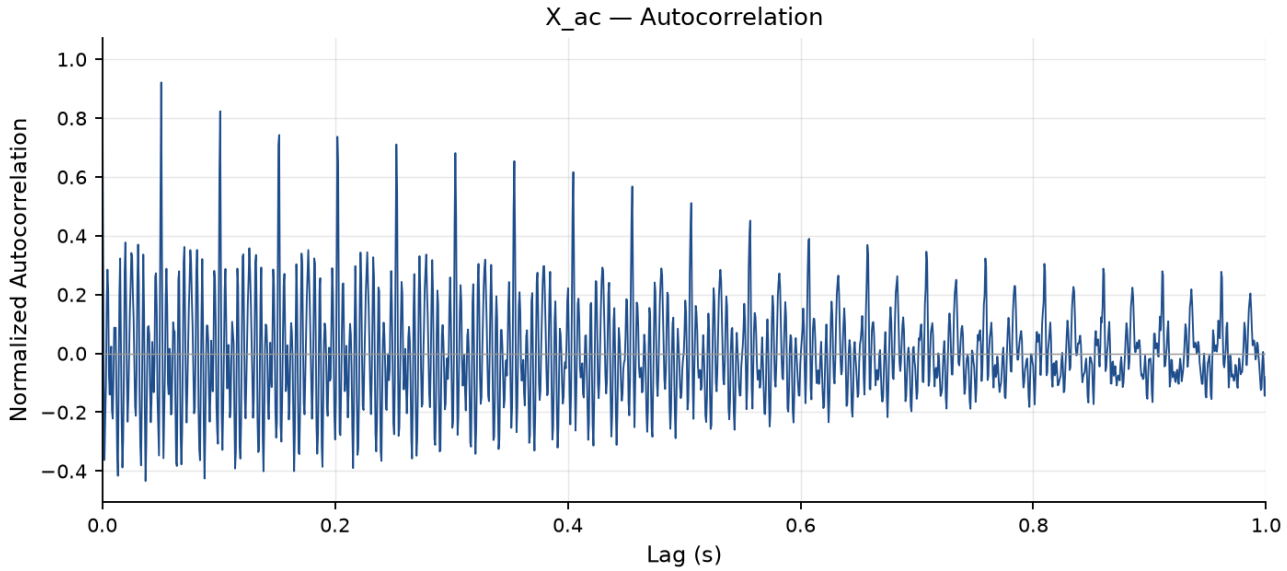
Z_ac — spectrogram.



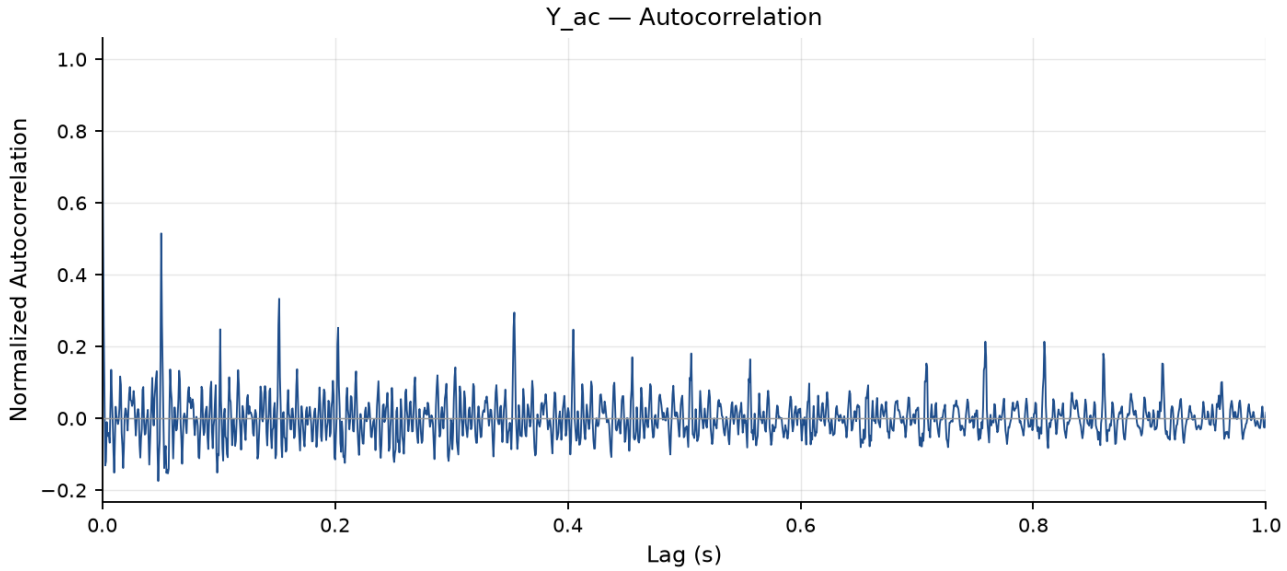
R_g — spectrogram.

10. Autocorrelation Analysis

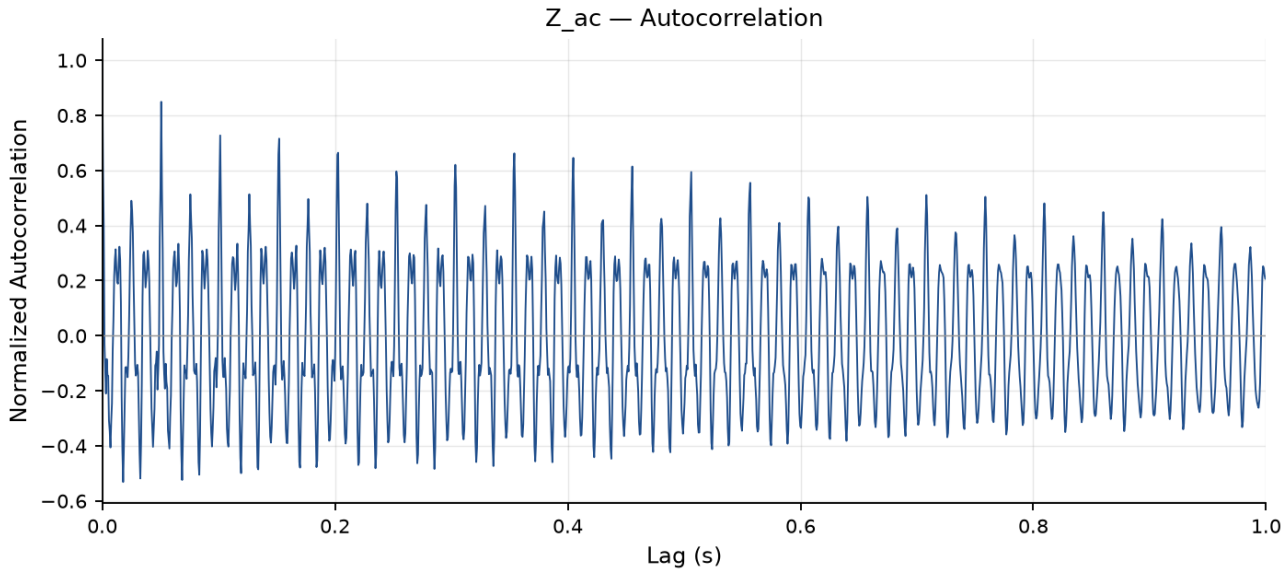
Normalized autocorrelation up to a 1.0 s lag, for visual inspection of periodicity.



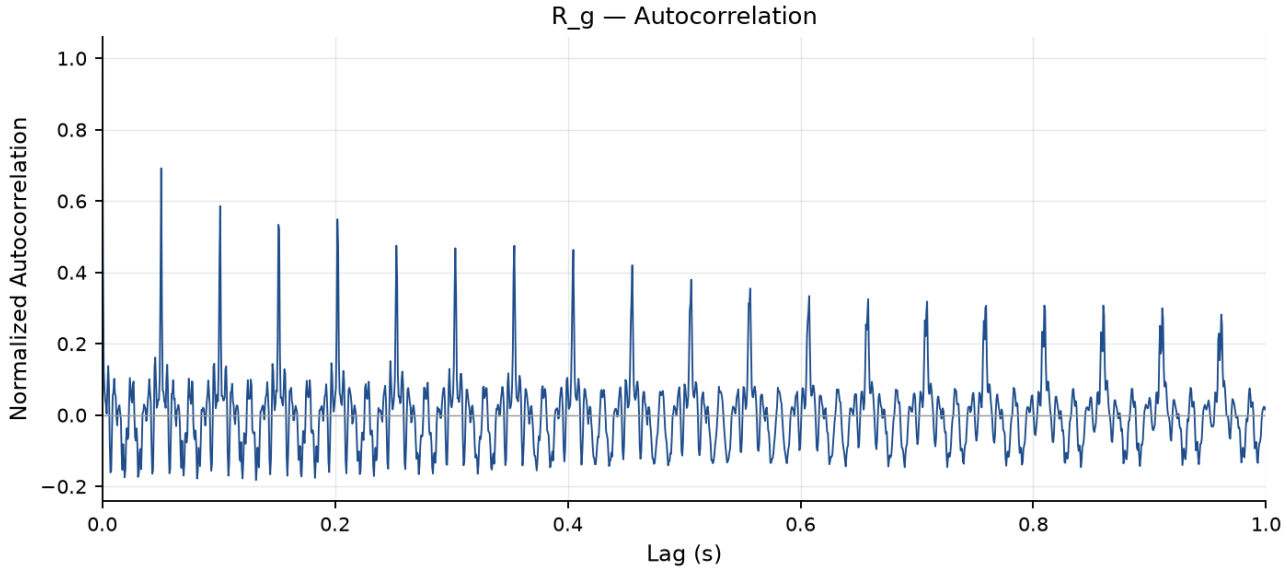
X_ac — autocorrelation.



Y_ac — autocorrelation.



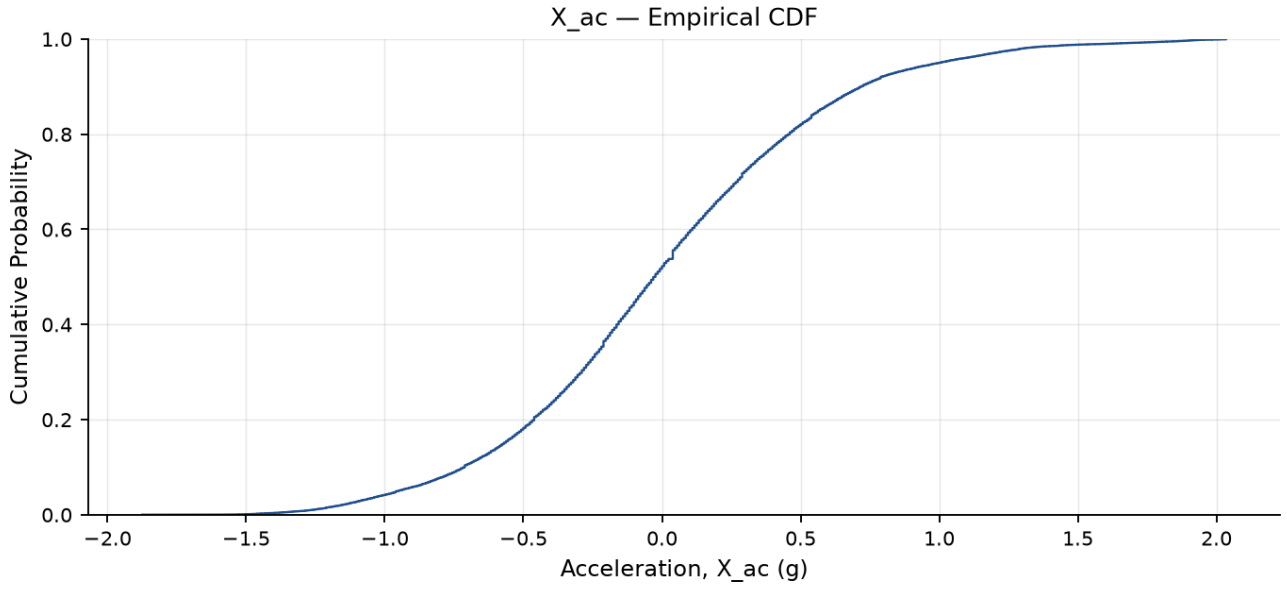
Z_ac — autocorrelation.



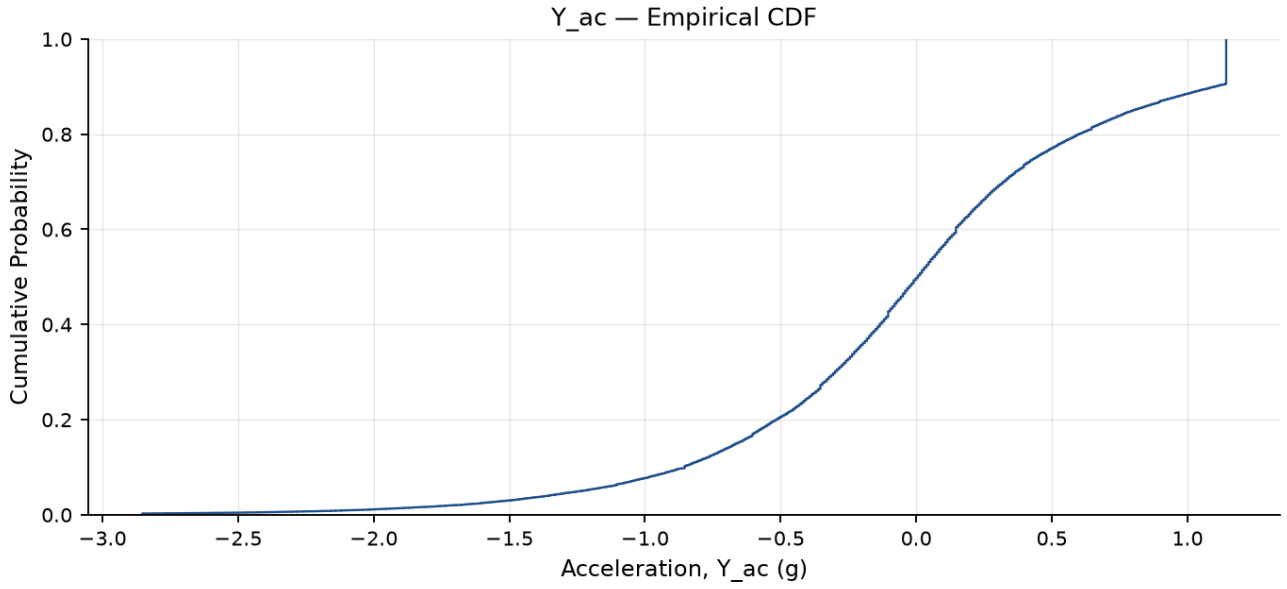
R_g — autocorrelation.

11. Empirical CDF

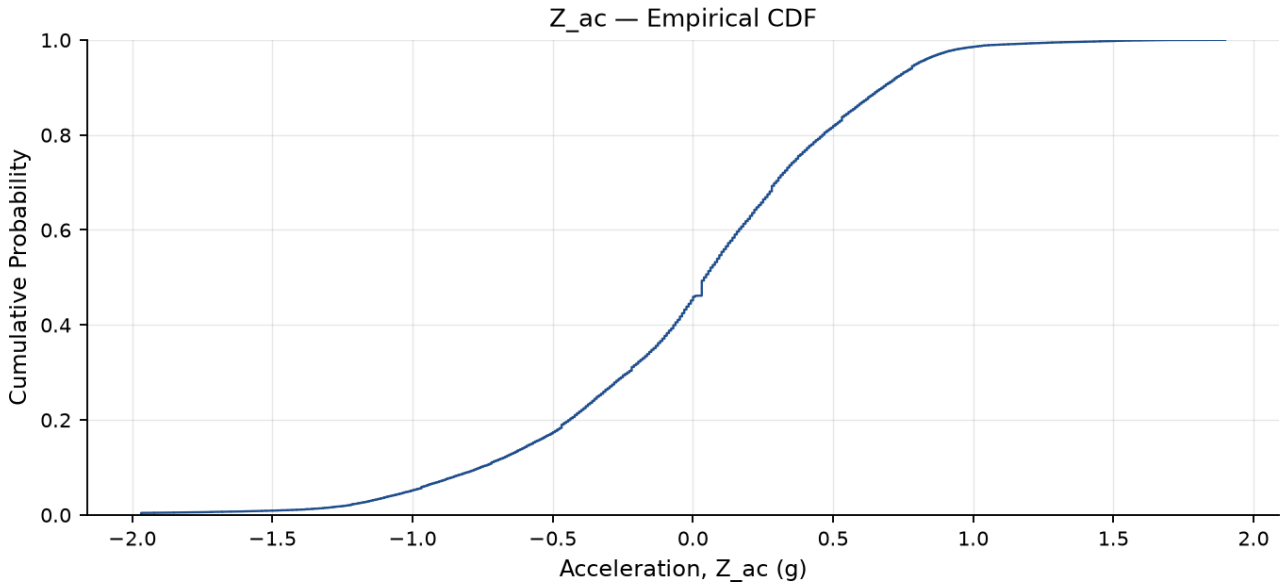
Empirical cumulative distribution functions computed directly from the measured data (no assumed distribution fitted).



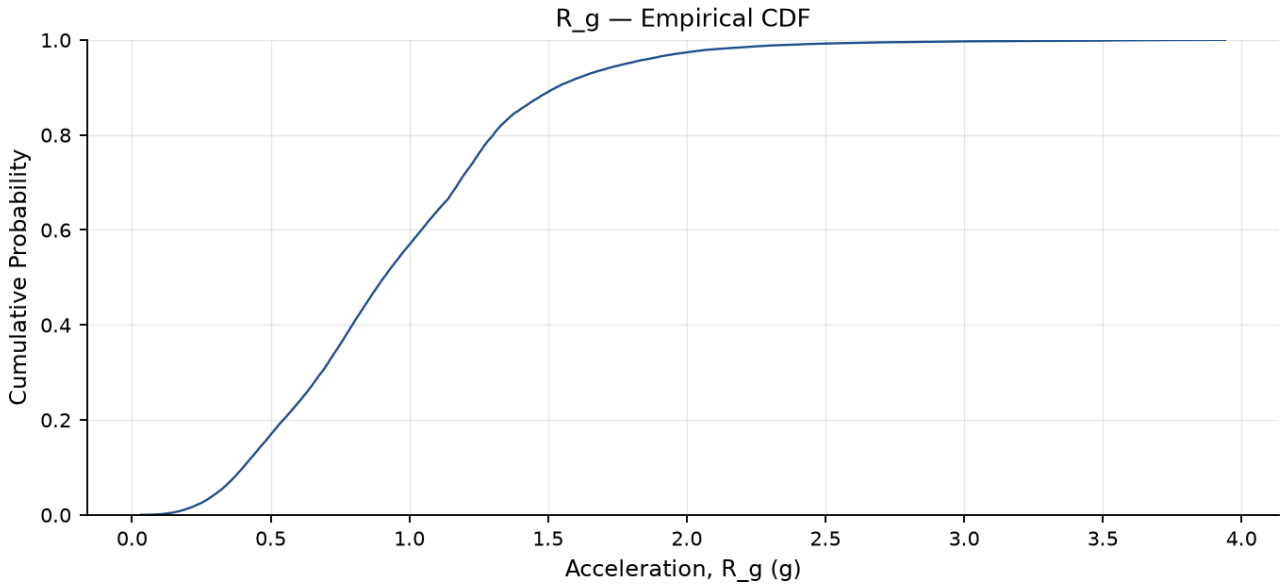
X_ac — empirical CDF.



Y_ac — empirical CDF.



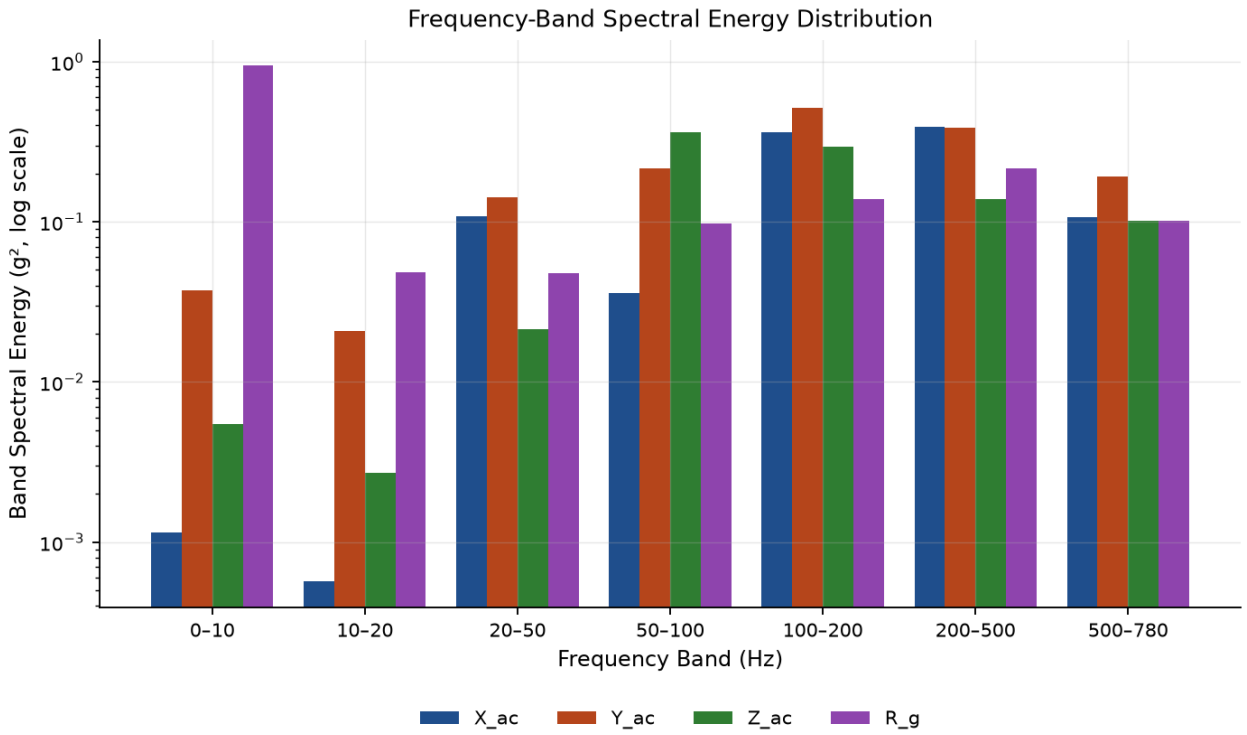
Z_ac — empirical CDF.



R_g — empirical CDF.

12. Spectral Energy

Spectral energy distributed across fixed visualization frequency bands (final band extended to the actual Nyquist frequency, 779.5 Hz). Bands are for visualization only and were not used as an analysis boundary elsewhere.



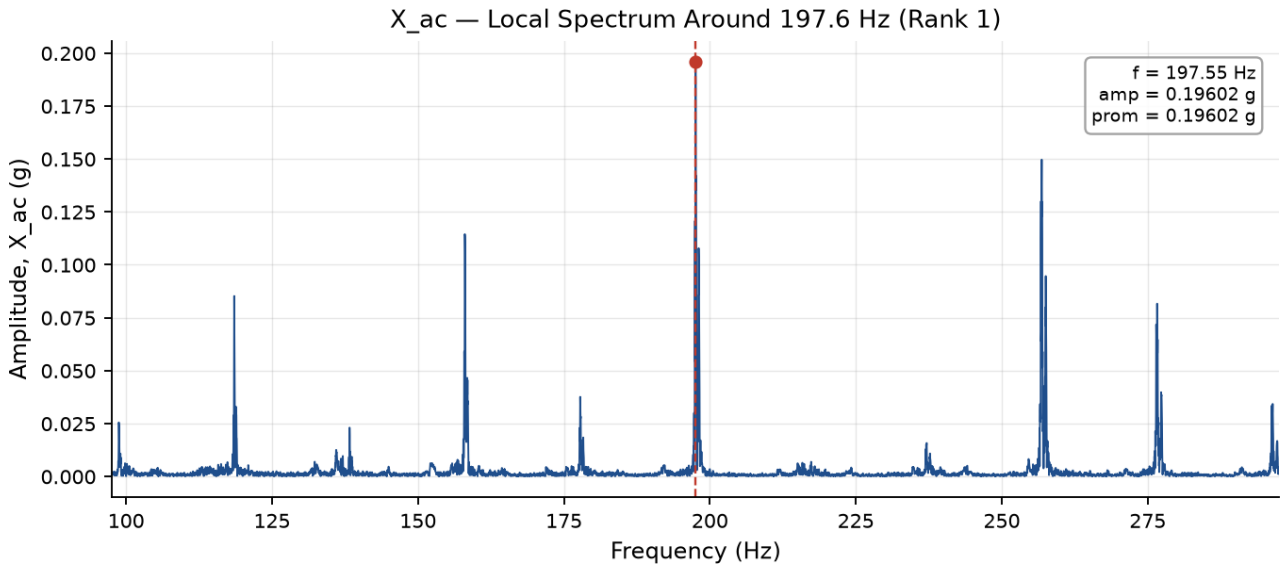
Frequency-band spectral energy, all signals.

Band (Hz)	X_ac	Y_ac	Z_ac	R_g
0–10.0	0.0011486	0.037193	0.0054438	0.94692
10–20.0	5.6632e-04	0.020832	0.0027215	0.048398
20–50.0	0.10827	0.14179	0.021355	0.047806
50–100.0	0.035926	0.2167	0.36435	0.097783
100–200.0	0.36112	0.5155	0.29315	0.13852
200–500.0	0.39265	0.38861	0.13777	0.21639
500–779.5	0.10714	0.19246	0.10206	0.10155

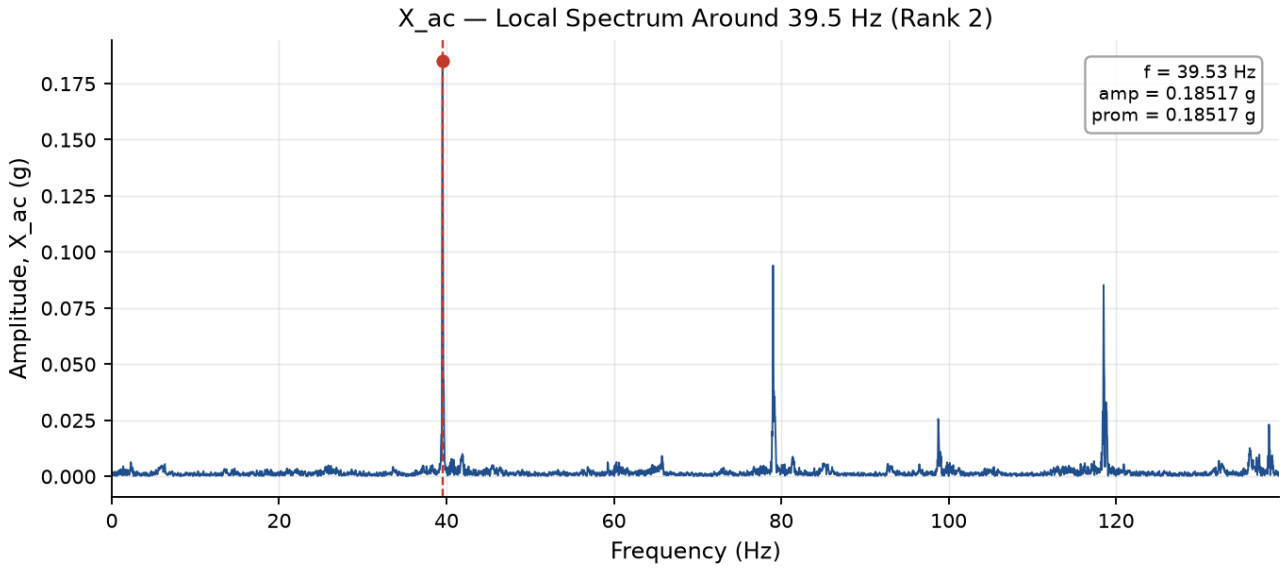
13. Local Spectral Analysis (Automatic ±100 Hz Zoom)

For the strongest automatically detected significant peaks in each signal's full spectrum (adaptive prominence, see Section 6/methodology), a local ±100 Hz window is shown with a compact annotation (frequency, amplitude, prominence). Peaks are ranked by prominence; up to 5 peaks are shown per signal. "Equivalent RPM" values, where given, denote frequency × 60 and describe only what would follow IF that component corresponded to 1× shaft rotation — no such correspondence is asserted by this report. Detailed peak data for all signals is saved to spectral_peaks.csv and local_spectral_analysis.csv for further analysis.

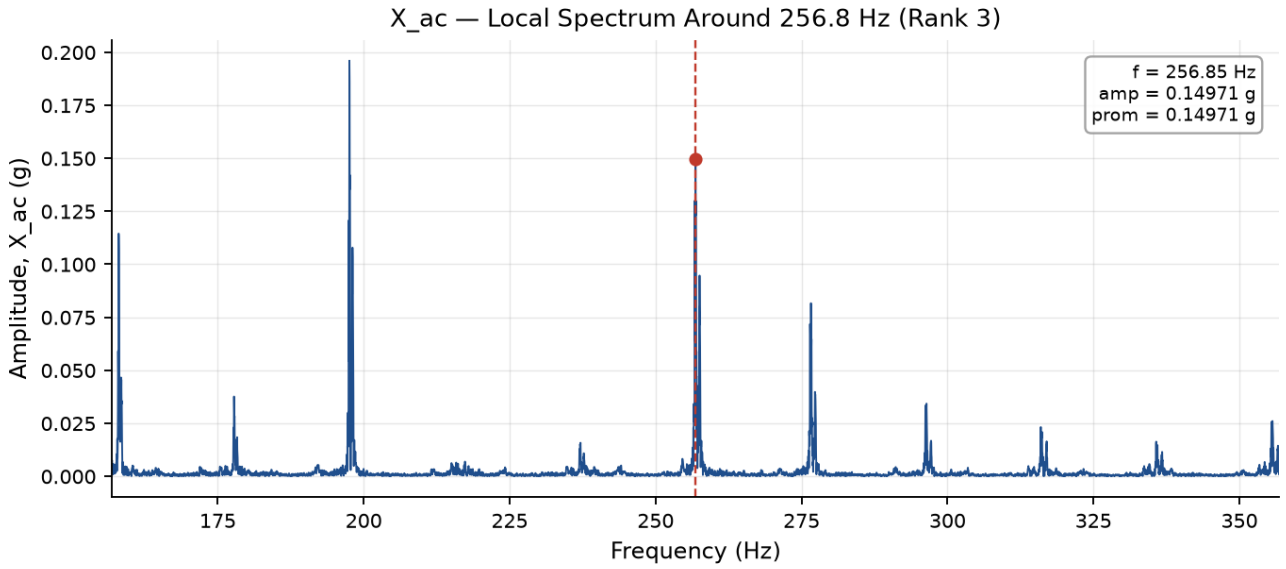
X_ac — Significant Peaks



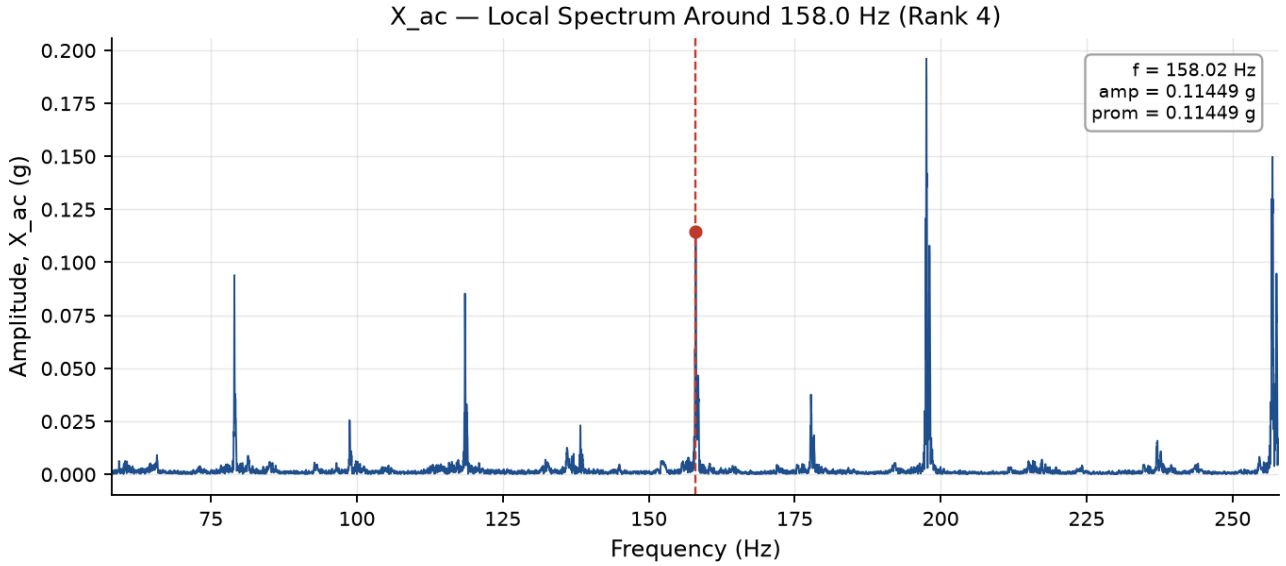
X_ac rank 1: f = 197.55 Hz, amplitude = 0.196 g, prominence = 0.196 g (equivalent RPM if 1×: 11853).



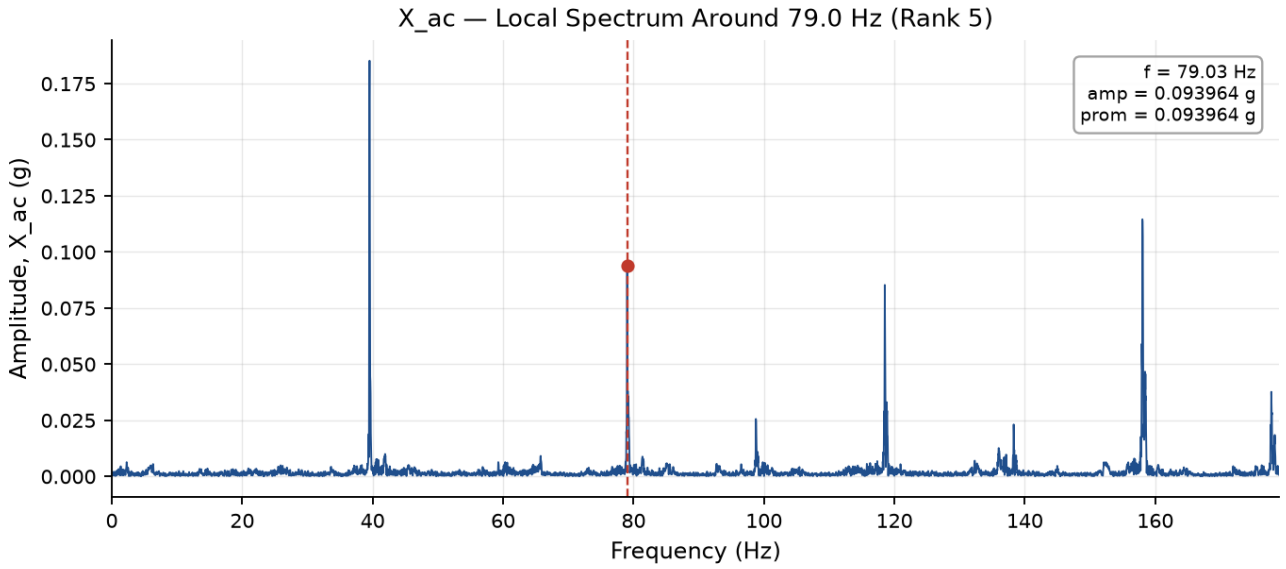
X_ac rank 2: f = 39.53 Hz, amplitude = 0.1852 g, prominence = 0.1852 g (equivalent RPM if 1×: 2372).



X_ac rank 3: f = 256.85 Hz, amplitude = 0.1497 g, prominence = 0.1497 g (equivalent RPM if 1×: 15411).

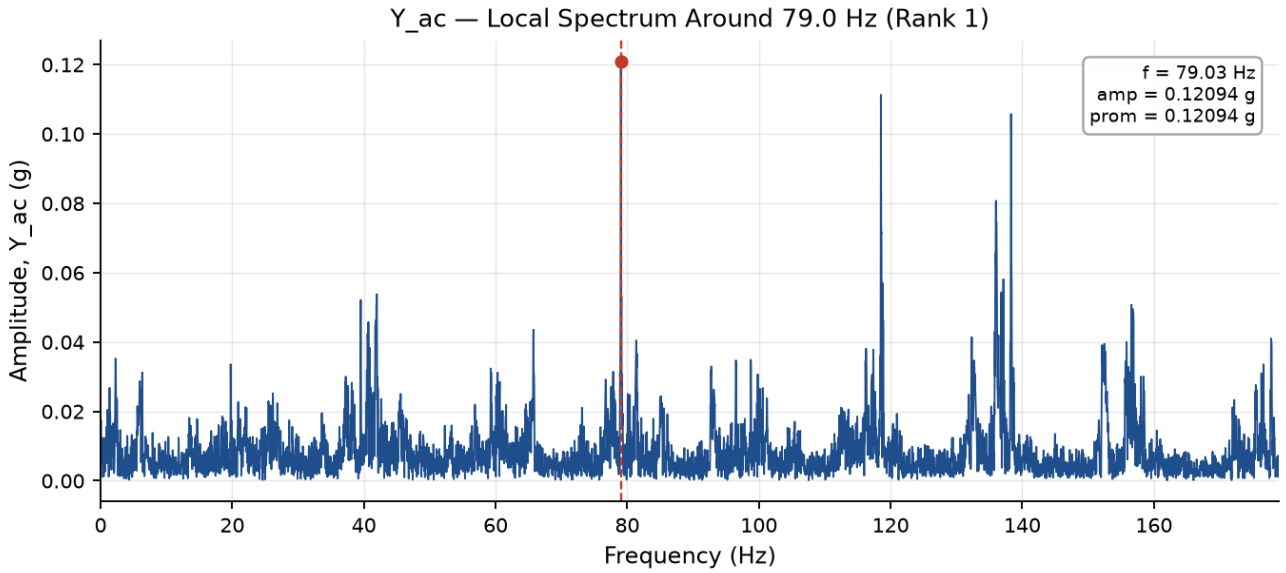


X_ac rank 4: f = 158.02 Hz, amplitude = 0.1145 g, prominence = 0.1145 g (equivalent RPM if 1×: 9481).

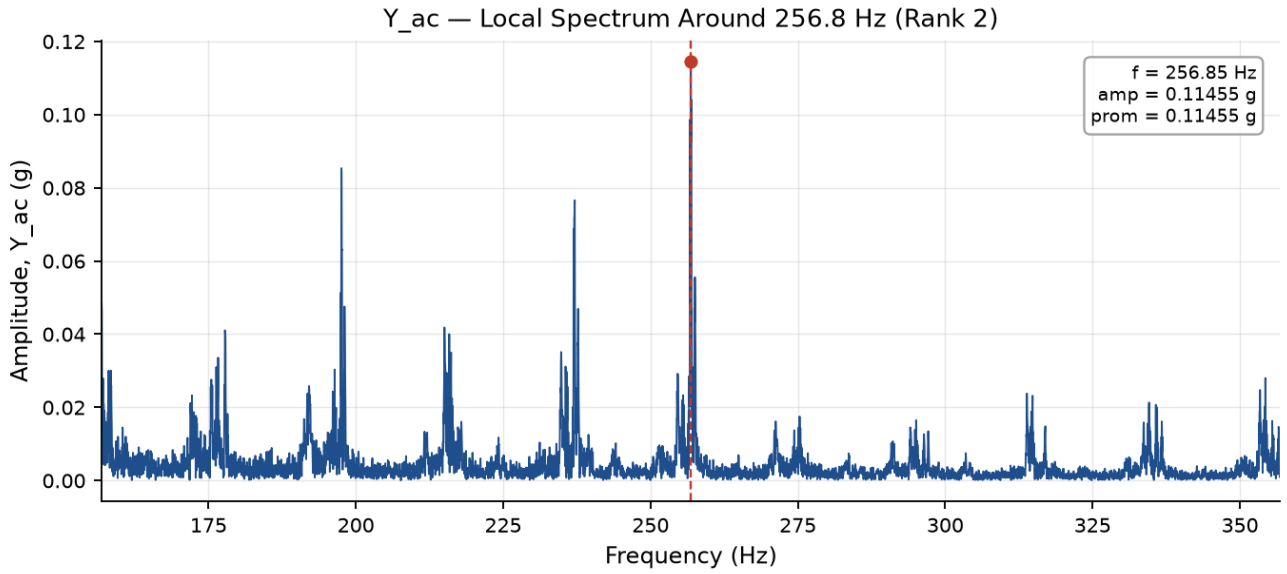


X_ac rank 5: f = 79.03 Hz, amplitude = 0.09396 g, prominence = 0.09396 g (equivalent RPM if 1×: 4742).

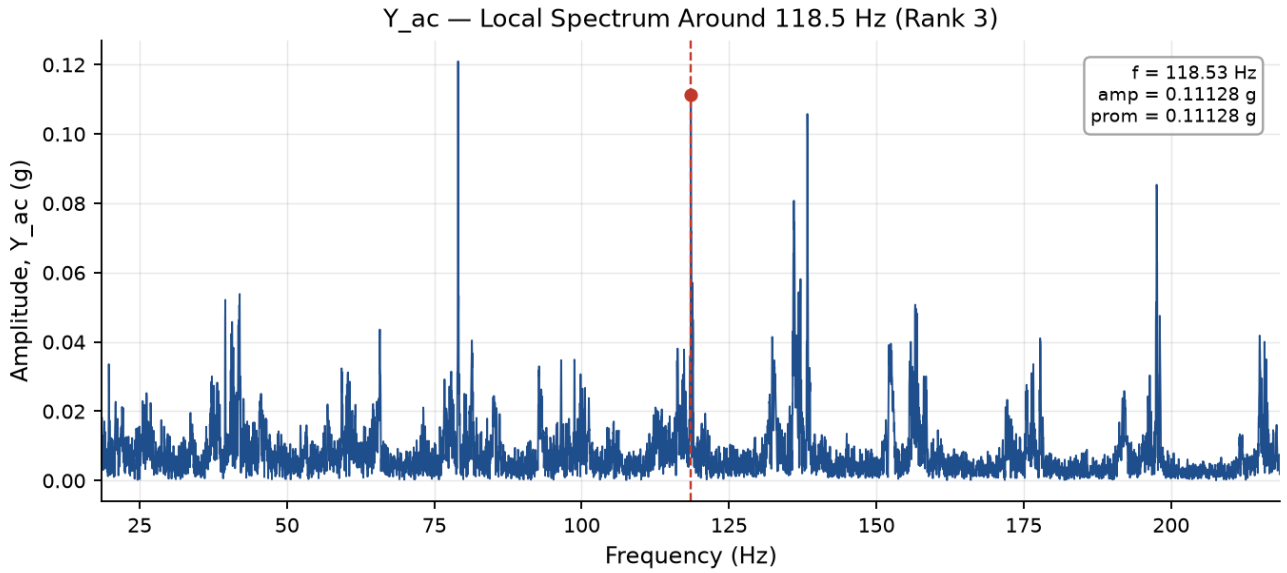
Y_ac — Significant Peaks



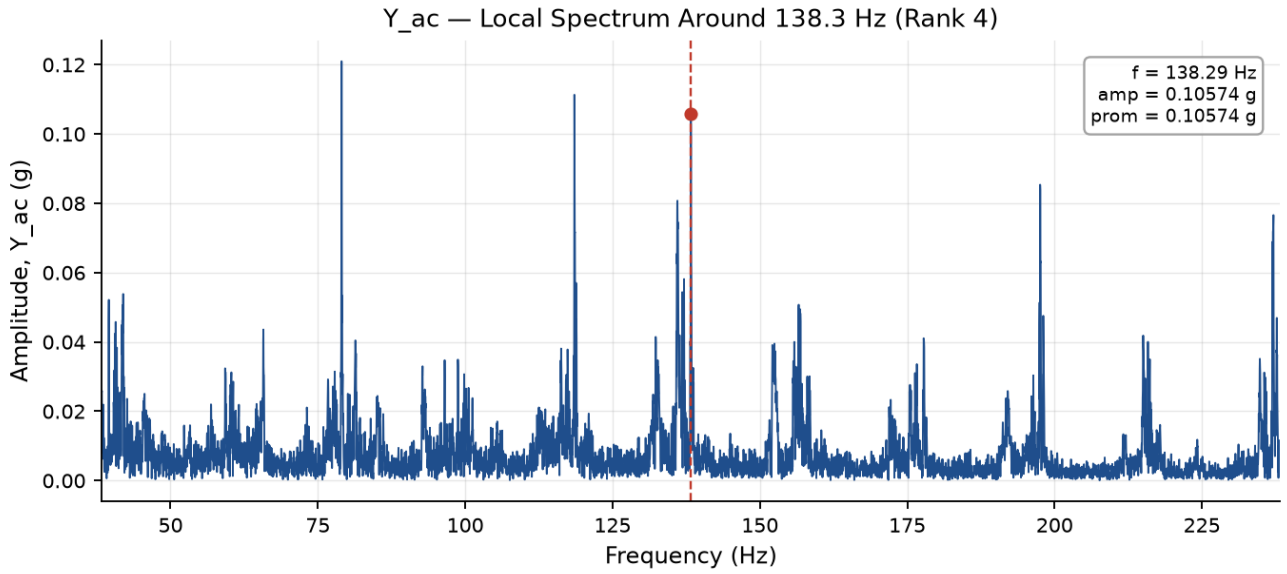
Y_ac rank 1: f = 79.03 Hz, amplitude = 0.1209 g, prominence = 0.1209 g (equivalent RPM if 1×: 4742).



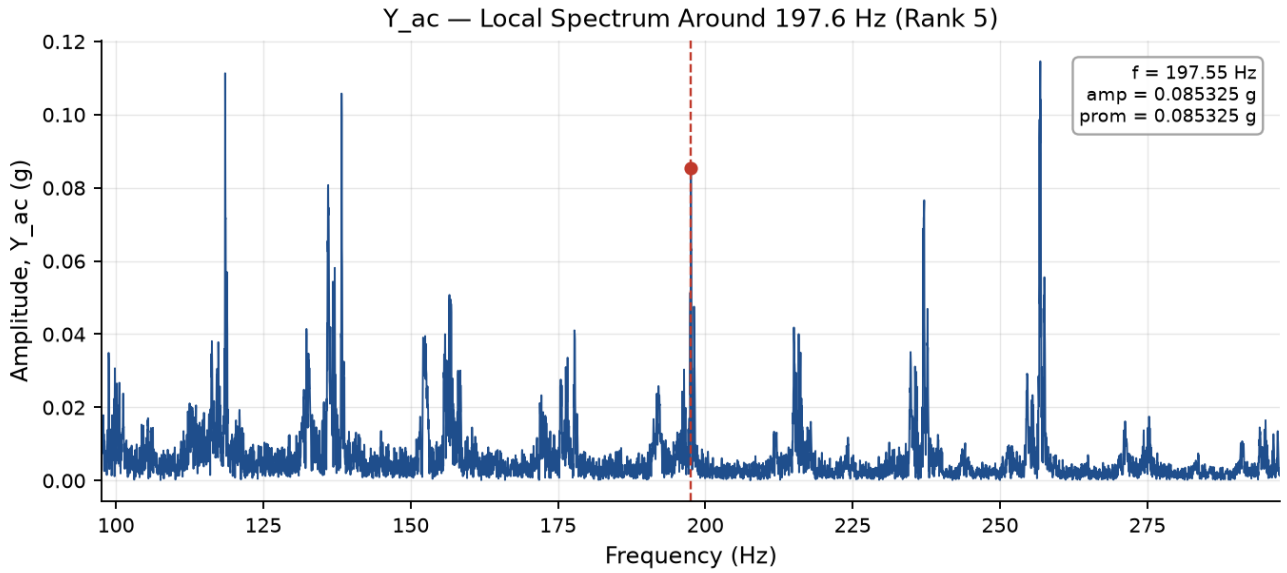
Y_ac rank 2: f = 256.85 Hz, amplitude = 0.1145 g, prominence = 0.1145 g (equivalent RPM if 1×: 15411).



Y_ac rank 3: f = 118.53 Hz, amplitude = 0.1113 g, prominence = 0.1113 g (equivalent RPM if 1×: 7112).

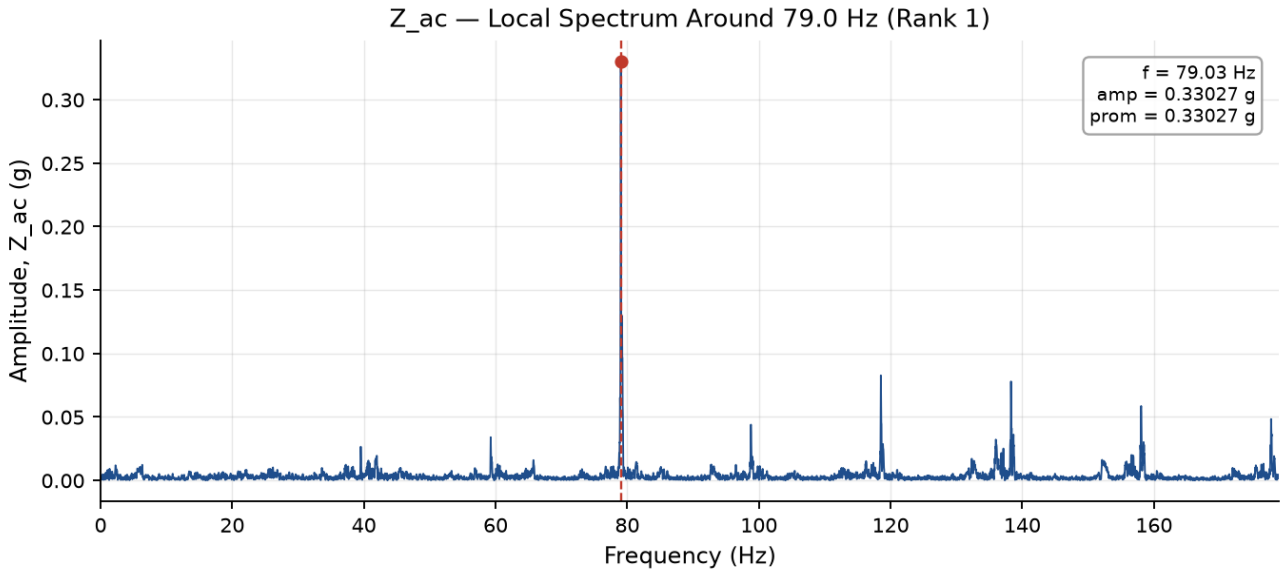


Y_ac rank 4: f = 138.29 Hz, amplitude = 0.1057 g, prominence = 0.1057 g (equivalent RPM if 1×: 8297).

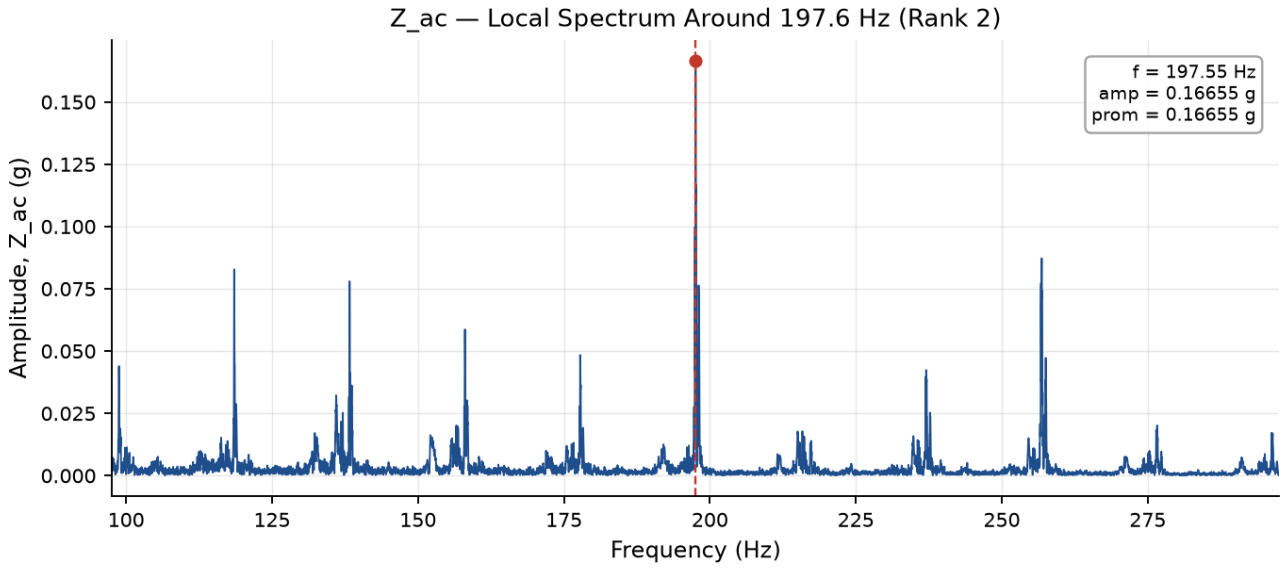


Y_ac rank 5: f = 197.55 Hz, amplitude = 0.08533 g, prominence = 0.08532 g (equivalent RPM if 1×: 11853).

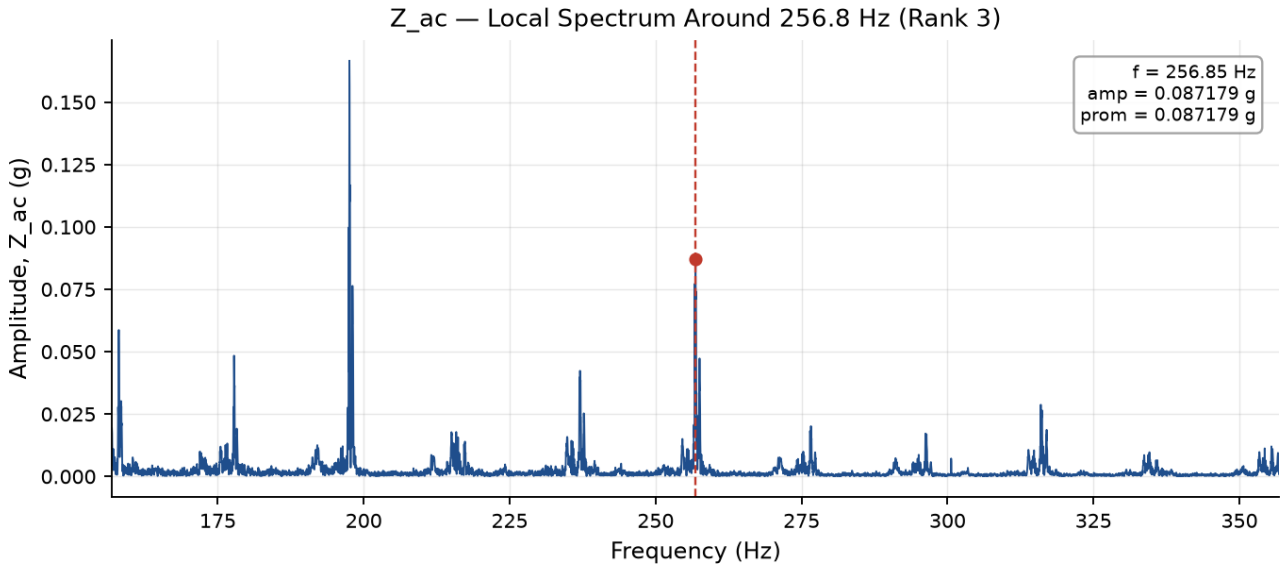
Z_ac — Significant Peaks



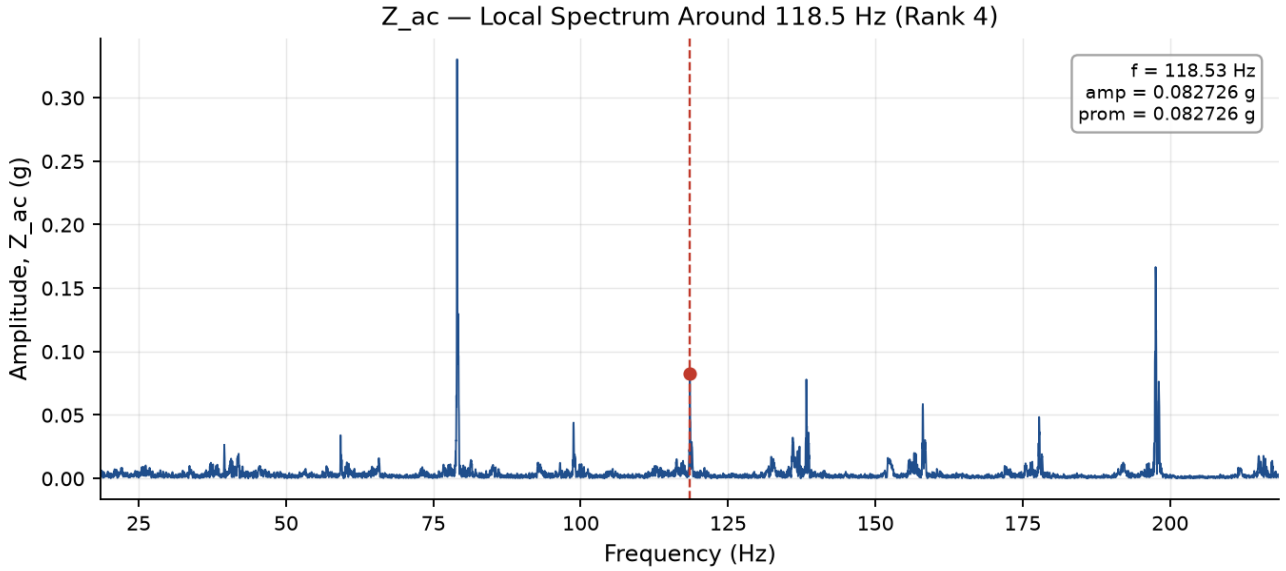
Z_ac rank 1: f = 79.03 Hz, amplitude = 0.3303 g, prominence = 0.3303 g (equivalent RPM if 1×: 4742).



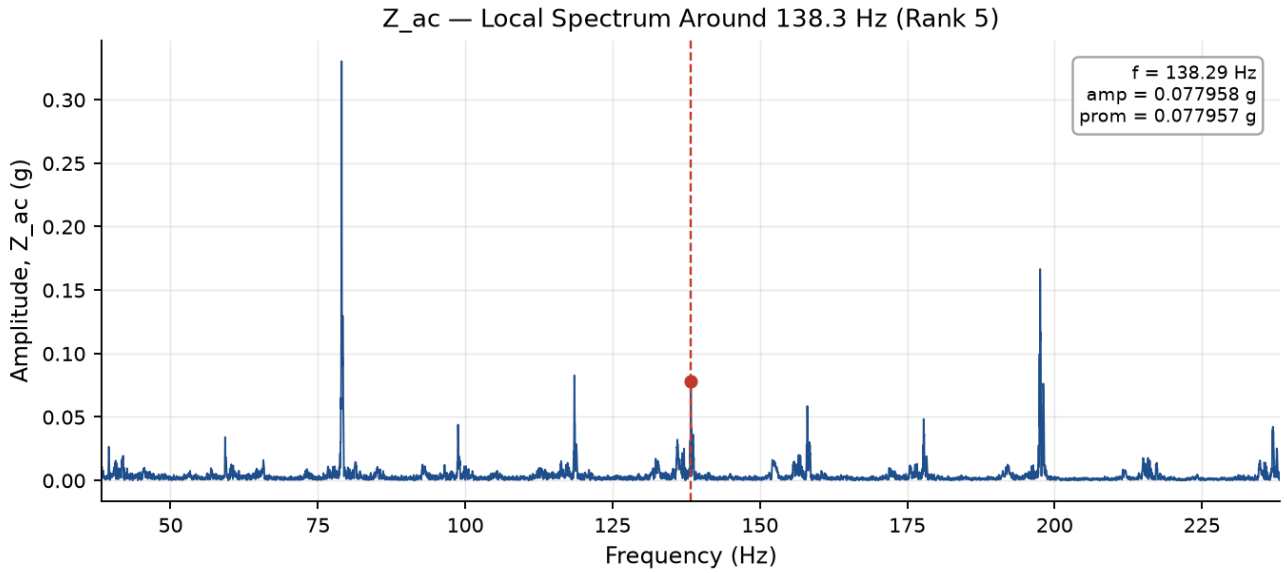
Z_ac rank 2: f = 197.55 Hz, amplitude = 0.1665 g, prominence = 0.1665 g (equivalent RPM if 1×: 11853).



Z_ac rank 3: f = 256.85 Hz, amplitude = 0.08718 g, prominence = 0.08718 g (equivalent RPM if 1×: 15411).

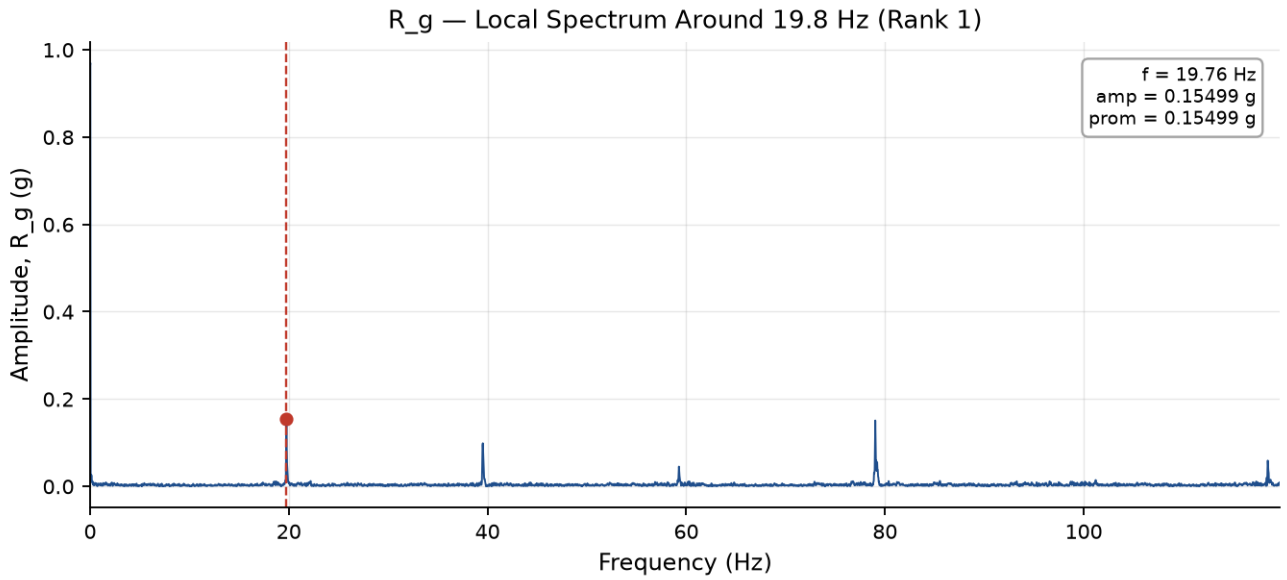


Z_ac rank 4: f = 118.53 Hz, amplitude = 0.08273 g, prominence = 0.08273 g (equivalent RPM if 1×: 7112).

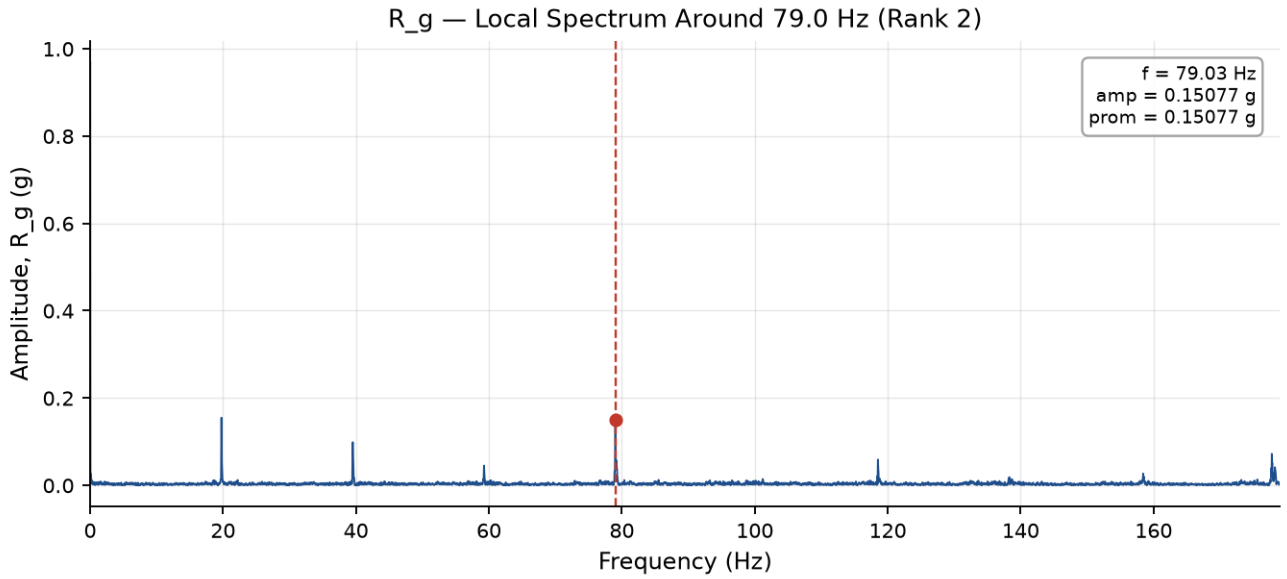


Z_ac rank 5: f = 138.29 Hz, amplitude = 0.07796 g, prominence = 0.07796 g (equivalent RPM if 1×: 8297).

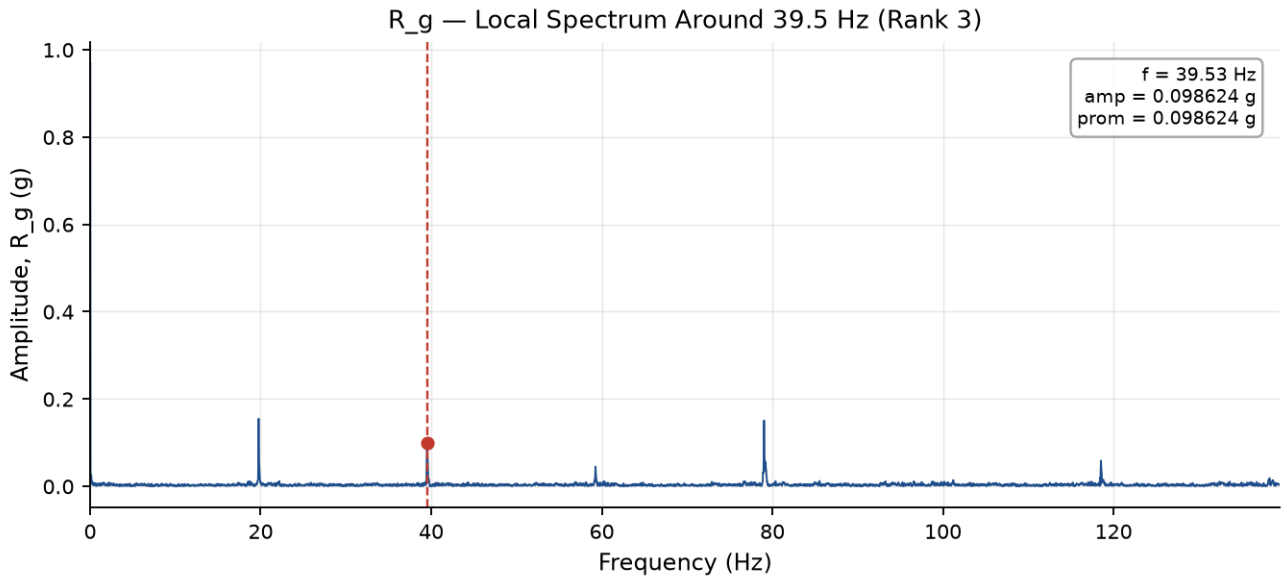
R_g — Significant Peaks



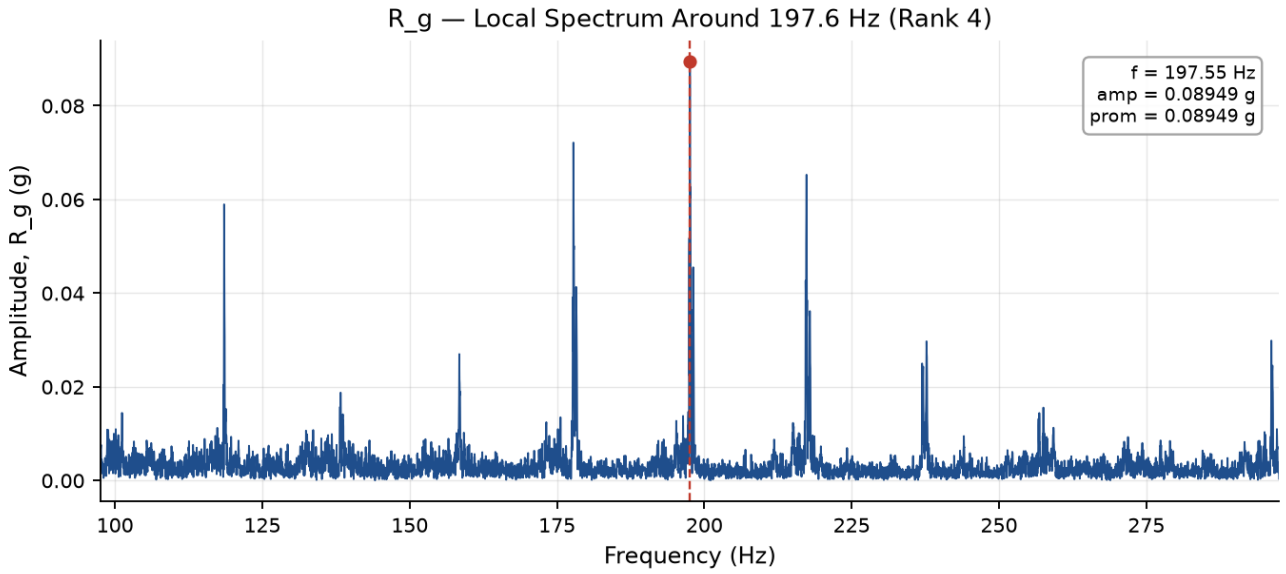
R_g rank 1: f = 19.76 Hz, amplitude = 0.155 g, prominence = 0.155 g (equivalent RPM if 1×: 1186).



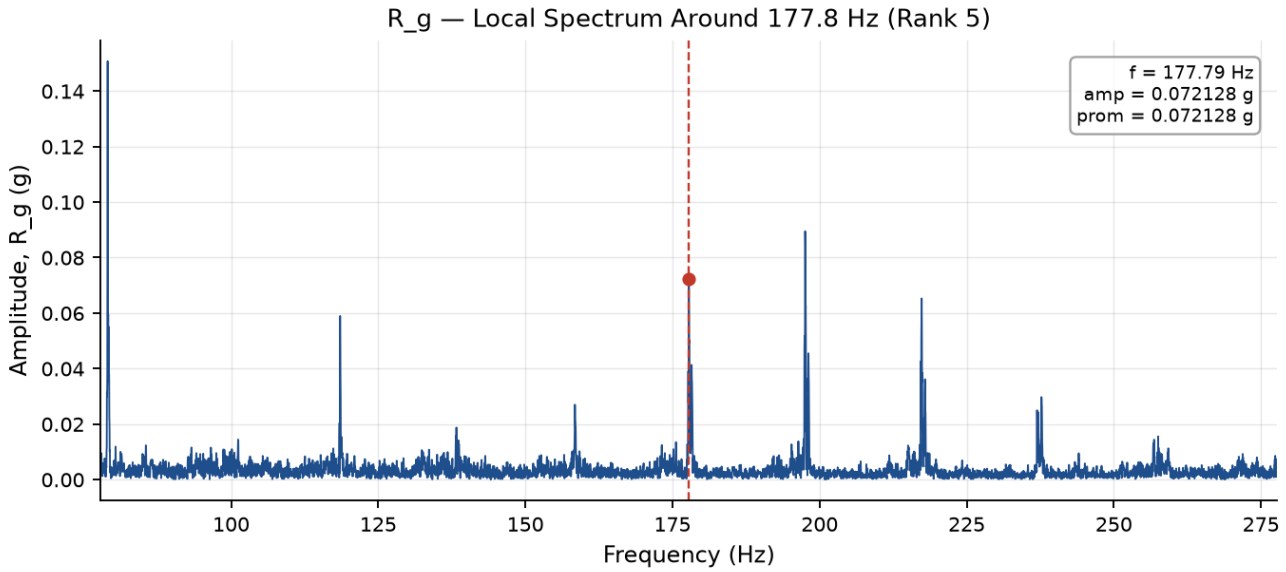
R_g rank 2: f = 79.03 Hz, amplitude = 0.1508 g, prominence = 0.1508 g (equivalent RPM if 1×: 4742).



R_g rank 3: f = 39.53 Hz, amplitude = 0.09862 g, prominence = 0.09862 g (equivalent RPM if 1×: 2372).



R_g rank 4: f = 197.55 Hz, amplitude = 0.08949 g, prominence = 0.08949 g (equivalent RPM if 1×: 11853).



R_g rank 5: f = 177.79 Hz, amplitude = 0.07213 g, prominence = 0.07213 g (equivalent RPM if 1×: 10667).